

STAT 423 Winter 2026 - Final Project

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Data Processing - Krish Doshi

```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating, age=orig_data$age,
                           gender = orig_data$gender, race = orig_data$race, genre = orig_data$genre, bpm=orig_data$bpm,
                           energy=orig_data$energy, danceability = orig_data$danceability, has_winds = orig_data$has_winds,
                           has_brass = orig_data$has_brass, has_percussion = orig_data$has_percussion, has_vocal_or_whistle = orig_data$has_vocal_or_whistle)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified", orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
                             ifelse(grepl("Country", orig_data$cdr_genre), "Country",
                                     ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
                                             ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
                                                     ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy, danceability = orig_data$danceability)

#Wind
cleaned_data$has_winds <- ifelse(orig_data$clarinet, "True",
                                ifelse(orig_data$flute_piccolo, "True",
                                        ifelse(orig_data$harmonica, "True",
                                              ifelse(orig_data$kazoo, "True",
                                                      ifelse(orig_data$ocarina, "True",
                                                                ifelse(orig_data$saxophone, "True",
                                                                      ifelse(orig_data$accordion, "True", "False"))))))))

#Brass
cleaned_data$has_brass <- ifelse(orig_data$trumpet, "True", "False")

#Percussion
cleaned_data$has_percussion <- ifelse(orig_data$piano_keyboard_based, "True",
                                      ifelse(orig_data$bongos, "True",
                                              ifelse(orig_data$cowbell, "True",
                                                      ifelse(orig_data$handclaps_snaps, "True", "False")))))

#Vocal or Whistle
cleaned_data$has_vocal_or_whistle <- ifelse(orig_data$vocally_based, "True",
```

```

        ifelse(orig_data$falsetto_vocal, "True",
              ifelse(orig_data$human_whistling, "True", "False")))

#String
cleaned_data$has_strings <- ifelse(orig_data$bass_based, "True",
                                ifelse(orig_data$guitar_based, "True",
                                      ifelse(orig_data$orchestral_strings, "True",
                                            ifelse(orig_data$banjo, "True",
                                                  ifelse(orig_data$mandolin, "True",
                                                        ifelse(orig_data$pedal_lap_steel, "True",
                                                                ifelse(orig_data$sitar, "True",
                                                                    ifelse(orig_data$ukulele, "True",
                                                                        ifelse(orig_data$violin, "True", "False")))))))

cleaned_data$year <- as.numeric(substr(orig_data$date, 1, unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]

cutoff <- round(min(cleaned_data$year) + ((max(cleaned_data$year) - min(cleaned_data$year)) / 2))

first_subpop <- cleaned_data[cleaned_data$year < cutoff, ]

second_subpop <- cleaned_data[cleaned_data$year >= cutoff, ]

```

Models - Kian Sharifzadeh

```

model_subpop1_1 <- lm(data = first_subpop, rating ~ age + race + gender + genre + energy + danceability
summary(model_subpop1_1)

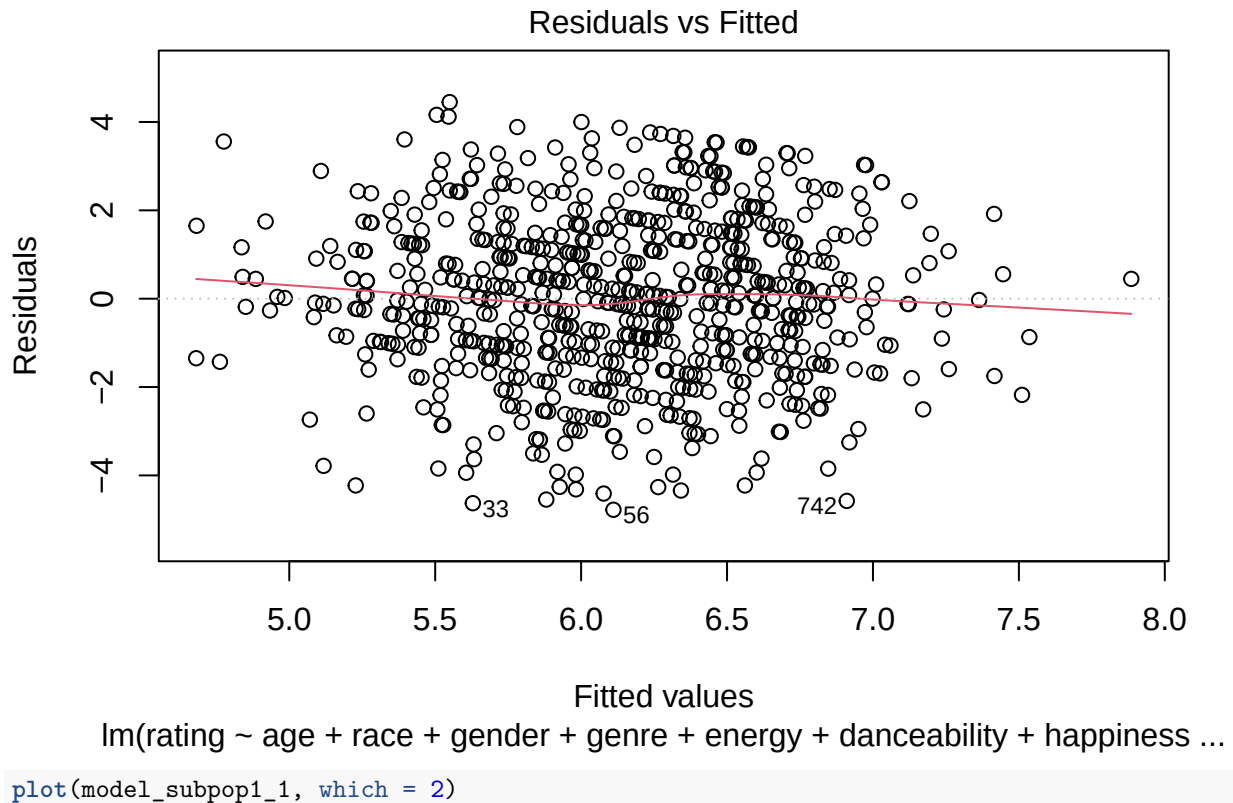
```

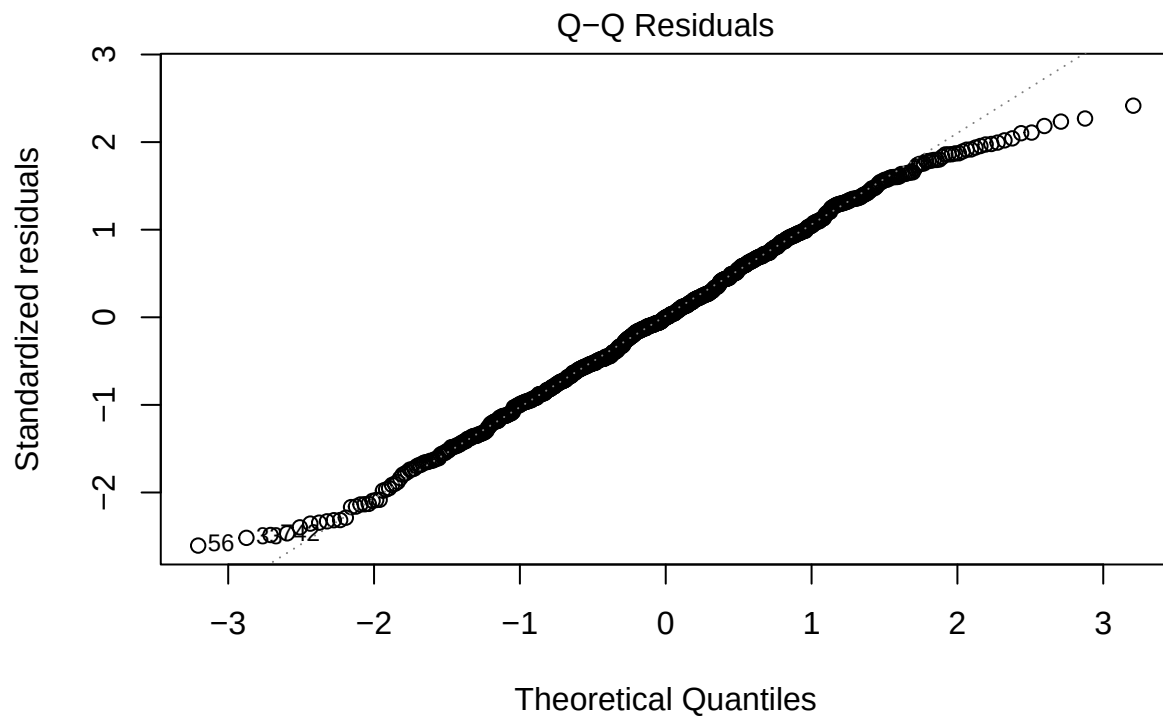
```

##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##     danceability + happiness + bpm + loudness_db + length_sec,
##     data = first_subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7774 -1.2578 -0.0063  1.3111  4.4504
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.012405   0.825236   9.709 < 2e-16 ***
## age          -0.030320   0.010595  -2.862  0.00433 **
## raceWhite    -0.648978   0.163236  -3.976  7.72e-05 ***
## genderNot Male  0.001685   0.155477   0.011  0.99135
## genreJazz      1.046241   0.852095   1.228  0.21990
## genreOther     0.463522   0.413876   1.120  0.26310
## genrePop      -0.180256   0.386269  -0.467  0.64088
## genreRock      0.526001   0.394097   1.335  0.18239
## energy        -0.008542   0.005863  -1.457  0.14555
## danceability  -0.013090   0.005734  -2.283  0.02271 *
## happiness      0.006972   0.003926   1.776  0.07620 .

```

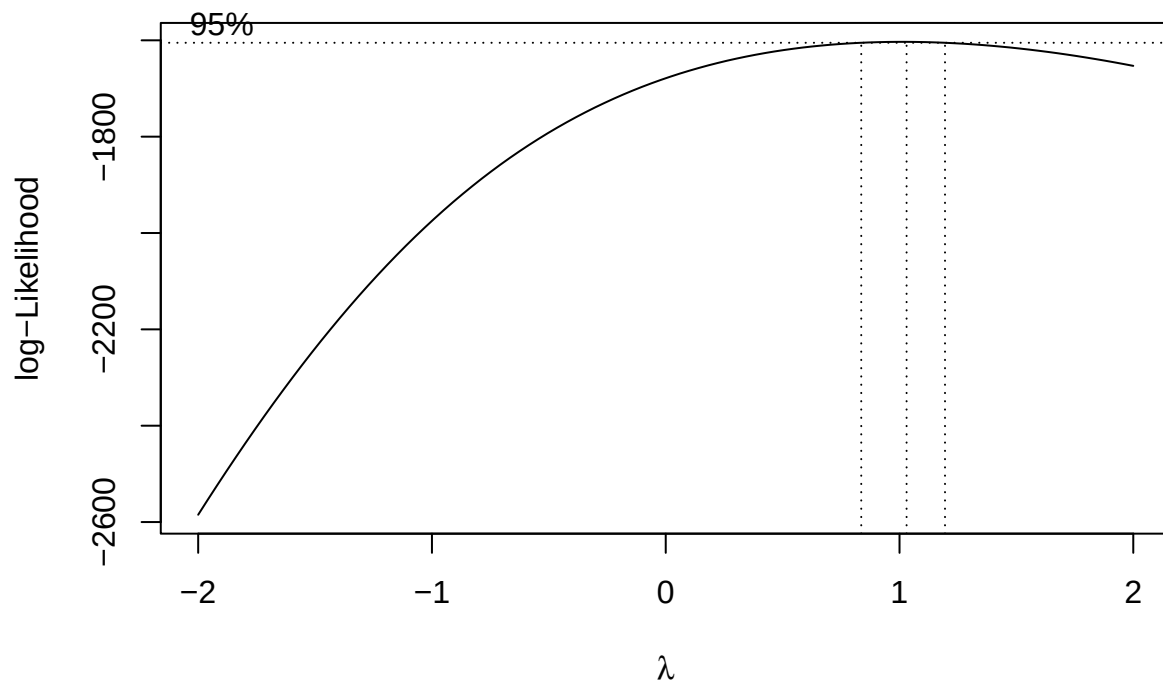
```
## bpm          0.001554  0.002986  0.520  0.60295
## loudness_db  0.063151  0.030448  2.074  0.03842 *
## length_sec   0.002120  0.001475  1.437  0.15111
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.856 on 729 degrees of freedom
## Multiple R-squared:  0.07473,    Adjusted R-squared:  0.05823
## F-statistic: 4.529 on 13 and 729 DF,  p-value: 1.76e-07
#residual anaylsis plots for subpop 1
plot(model_subpop1_1, which = 1)
```





lm(rating ~ age + race + gender + genre + energy + danceability + happiness ...

```
boxcox(model_subpop1_1, plotit=T)
```



```
step(model_subpop1_1, direction="backward")
```

```
## Start:  AIC=932.67
## rating ~ age + race + gender + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec
##
##           Df Sum of Sq  RSS   AIC
```

```

## - gender      1      0.000 2510.6 930.67
## - bpm         1      0.933 2511.6 930.95
## <none>                2510.6 932.67
## - length_sec  1      7.113 2517.7 932.77
## - energy       1      7.311 2517.9 932.83
## - happiness    1     10.859 2521.5 933.88
## - loudness_db  1     14.815 2525.4 935.04
## - danceability 1     17.952 2528.6 935.96
## - age          1     28.205 2538.8 938.97
## - genre        4     65.376 2576.0 943.77
## - race         1     54.435 2565.1 946.61
##
## Step: AIC=930.67
## rating ~ age + race + genre + energy + danceability + happiness +
##      bpm + loudness_db + length_sec
##
##           Df Sum of Sq   RSS   AIC
## - bpm      1      0.933 2511.6 928.95
## <none>                2510.6 930.67
## - length_sec  1      7.197 2517.8 930.80
## - energy      1      7.315 2517.9 930.83
## - happiness    1     10.900 2521.5 931.89
## - loudness_db  1     14.822 2525.4 933.04
## - danceability 1     17.984 2528.6 933.97
## - age         1     28.574 2539.2 937.08
## - genre       4     70.110 2580.7 943.13
## - race        1     55.065 2565.7 944.79
##
## Step: AIC=928.95
## rating ~ age + race + genre + energy + danceability + happiness +
##      loudness_db + length_sec
##
##           Df Sum of Sq   RSS   AIC
## <none>                2511.6 928.95
## - energy      1      6.780 2518.3 928.95
## - length_sec  1      7.263 2518.8 929.09
## - happiness    1     10.872 2522.4 930.15
## - loudness_db  1     14.554 2526.1 931.24
## - danceability 1     18.112 2529.7 932.28
## - age         1     28.643 2540.2 935.37
## - genre       4     69.691 2581.2 941.28
## - race        1     55.566 2567.1 943.20
##
## Call:
## lm(formula = rating ~ age + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec, data = first_subpop)
##
## Coefficients:
## (Intercept)          age      raceWhite      genreJazz      genreOther
##      8.177983     -0.030368     -0.651790       1.032634       0.442906
##      genrePop      genreRock          energy  danceability      happiness
##     -0.187951       0.520393     -0.008158     -0.013132       0.006959
##      loudness_db      length_sec

```

```
##      0.062536      0.002131
new_model_1 <- lm(data = cleaned_data, log(rating) ~ age+ gender+ race + log(bpm) + genre*log(energy) +
summary(new_model_1)

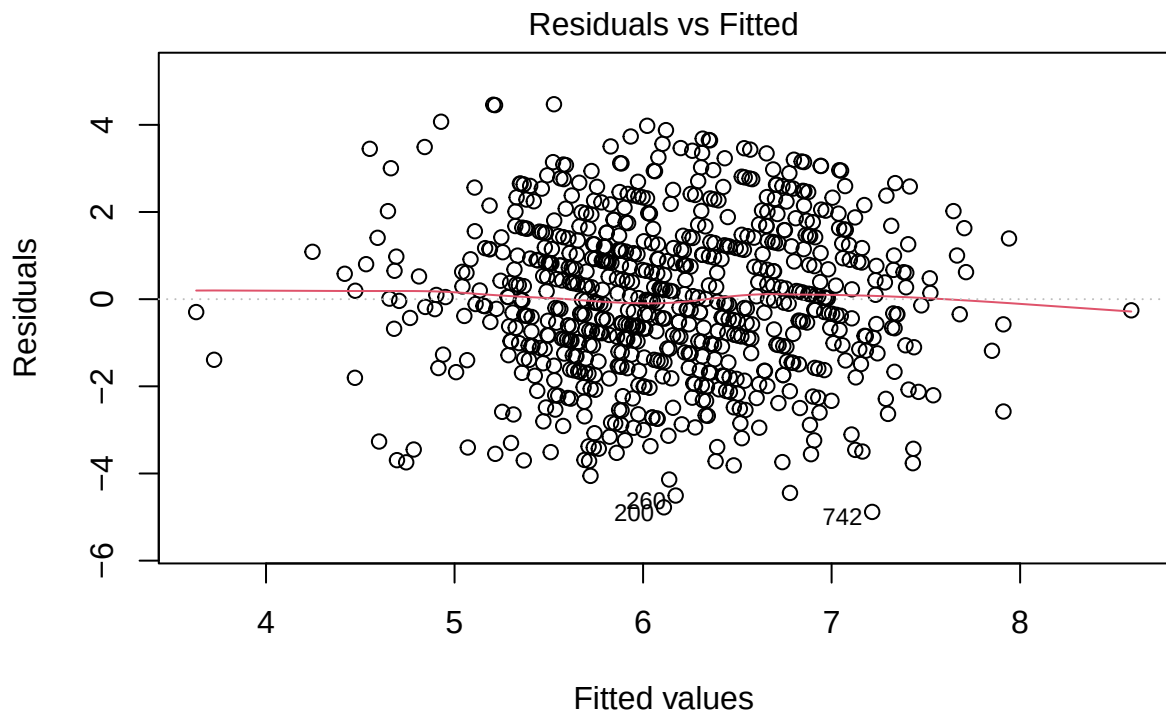
##
## Call:
## lm(formula = log(rating) ~ age + gender + race + log(bpm) + genre *
##     log(energy) + genre * log(danceability) + happiness + length_sec +
##     loudness_db + has_brass + has_percussion + has_vocal_or_whistle +
##     year * log(energy) + age * genre, data = cleaned_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.68362 -0.16641  0.04248  0.23600  0.66231
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.579e+01  1.399e+01  -1.128  0.259480
## age           -1.886e-03  1.057e-02  -0.178  0.858454
## genderNot Male    6.133e-02  2.143e-02   2.862  0.004289 **
## raceWhite      -8.932e-02  2.295e-02  -3.892  0.000105 ***
## log(bpm)        1.922e-02  4.546e-02   0.423  0.672501
## genreJazz       8.513e+00  1.103e+01   0.772  0.440345
## genreOther     -2.536e+00  9.507e-01  -2.668  0.007748 **
## genrePop       -2.312e+00  9.050e-01  -2.555  0.010746 *
## genreRock      -1.424e+00  9.660e-01  -1.474  0.140710
## log(energy)      6.332e+00  3.418e+00   1.852  0.064218 .
## log(danceability) -5.094e-01  2.100e-01  -2.426  0.015432 *
## happiness       1.145e-03  5.617e-04   2.039  0.041661 *
## length_sec      7.151e-04  2.211e-04   3.234  0.001256 **
## loudness_db     1.844e-02  4.770e-03   3.865  0.000117 ***
## has_brassTrue    6.642e-02  8.360e-02   0.795  0.427070
## has_percussionTrue -1.227e-02  2.067e-02  -0.594  0.552859
## has_vocal_or_whistleTrue 1.036e-01  2.377e-02   4.359  1.43e-05 ***
## year           1.022e-02  7.096e-03   1.440  0.150168
## genreJazz:log(energy) -1.507e+00  1.856e+00  -0.812  0.416936
## genreOther:log(energy) 1.163e-01  1.459e-01   0.797  0.425414
## genrePop:log(energy)  9.679e-02  1.342e-01   0.721  0.471028
## genreRock:log(energy) 1.463e-01  1.467e-01   0.997  0.318903
## genreJazz:log(danceability) -3.806e-01  8.979e-01  -0.424  0.671738
## genreOther:log(danceability) 5.201e-01  2.225e-01   2.338  0.019580 *
## genrePop:log(danceability) 5.170e-01  2.205e-01   2.345  0.019216 *
## genreRock:log(danceability) 2.685e-01  2.274e-01   1.180  0.238105
## log(energy):year -3.305e-03  1.739e-03  -1.900  0.057665 .
## age:genreJazz    -1.975e-02  2.348e-02  -0.841  0.400454
## age:genreOther    4.892e-03  1.084e-02   0.451  0.651753
## age:genrePop     -2.649e-03  1.081e-02  -0.245  0.806507
## age:genreRock    -1.190e-03  1.107e-02  -0.108  0.914381
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3293 on 1123 degrees of freedom
## Multiple R-squared:  0.103, Adjusted R-squared:  0.07899
## F-statistic: 4.296 on 30 and 1123 DF, p-value: 2.619e-13
```

```
model_subpop1_2 <- lm(data = first_subpop, rating ~ age + gender + race + age*genre + age*energy + age*
summary(model_subpop1_2)
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##     energy + age * danceability + age * happiness + age * bpm +
##     age * loudness_db + age * length_sec + race * genre + race *
##     energy + race * danceability + race * happiness + race *
##     bpm + race * loudness_db + race * length_sec + +gender *
##     genre + gender * energy + gender * danceability + gender *
##     happiness + gender * bpm + gender * loudness_db + gender *
##     length_sec, data = first_subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.8819 -1.2115 -0.0254  1.2281  4.4727
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.122e+01  4.075e+00   2.753  0.00606 **
## age           -1.025e-01  1.180e-01  -0.868  0.38556
## genderNot Male    9.917e-02  1.661e+00   0.060  0.95241
## raceWhite       -2.046e+00  2.079e+00  -0.984  0.32545
## genreJazz       -2.095e-02  3.139e+00  -0.007  0.99468
## genreOther      -1.395e+00  2.554e+00  -0.546  0.58513
## genrePop        -1.555e+00  2.455e+00  -0.634  0.52657
## genreRock       -9.519e-01  2.531e+00  -0.376  0.70695
## energy          7.785e-03  2.797e-02   0.278  0.78085
## danceability    -6.778e-02  3.036e-02  -2.233  0.02588 *
## happiness       2.095e-02  1.934e-02   1.083  0.27903
## bpm            5.520e-03  1.455e-02   0.379  0.70452
## loudness_db     1.118e-01  1.460e-01   0.766  0.44420
## length_sec     -9.933e-04  7.830e-03  -0.127  0.89909
## age:genreJazz   -1.055e-02  7.959e-02  -0.133  0.89456
## age:genreOther    5.163e-02  6.936e-02   0.744  0.45691
## age:genrePop     8.418e-03  6.482e-02   0.130  0.89672
## age:genreRock   -2.577e-03  6.760e-02  -0.038  0.96960
## age:energy      -9.549e-04  9.949e-04  -0.960  0.33748
## age:danceability 1.781e-03  9.875e-04   1.803  0.07179 .
## age:happiness   -2.467e-04  6.395e-04  -0.386  0.69983
## age:bpm         9.001e-05  4.617e-04   0.195  0.84550
## age:loudness_db -6.034e-04  4.927e-03  -0.122  0.90256
## age:length_sec  5.662e-05  2.685e-04   0.211  0.83308
## raceWhite:genreJazz 1.873e+00  2.108e+00   0.889  0.37447
## raceWhite:genreOther 4.061e-01  1.469e+00   0.277  0.78220
## raceWhite:genrePop 1.540e+00  1.439e+00   1.070  0.28488
## raceWhite:genreRock 1.960e+00  1.479e+00   1.325  0.18546
## raceWhite:energy  1.533e-02  1.287e-02   1.191  0.23407
## raceWhite:danceability -5.342e-03  1.251e-02  -0.427  0.66952
## raceWhite:happiness -4.542e-03  8.705e-03  -0.522  0.60200
## raceWhite:bpm    -1.108e-02  6.886e-03  -1.609  0.10817
## raceWhite:loudness_db -2.007e-02  6.400e-02  -0.314  0.75390
## raceWhite:length_sec 4.570e-03  3.333e-03   1.371  0.17078
```

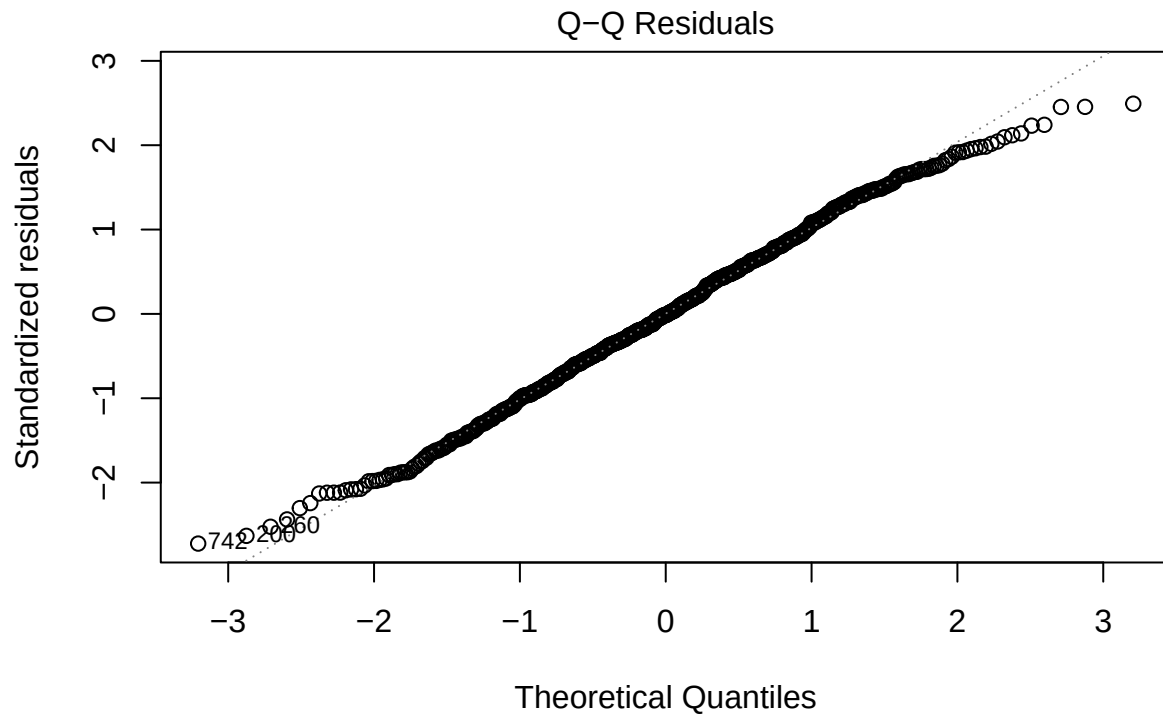
```
## genderNot Male:genreJazz          NA          NA          NA          NA
## genderNot Male:genreOther -1.664e+00  9.745e-01 -1.708  0.08810 .
## genderNot Male:genrePop -1.098e+00  9.203e-01 -1.193  0.23330
## genderNot Male:genreRock -1.412e+00  9.677e-01 -1.460  0.14487
## genderNot Male:energy 2.062e-03  1.381e-02  0.149  0.88132
## genderNot Male:danceability 3.385e-02  1.269e-02  2.667  0.00783 **
## genderNot Male:happiness -1.285e-02  8.571e-03 -1.499  0.13424
## genderNot Male:bpm 3.439e-03  6.717e-03  0.512  0.60886
## genderNot Male:loudness_db -4.889e-02  6.973e-02 -0.701  0.48346
## genderNot Male:length_sec -4.775e-03  3.415e-03 -1.398  0.16246
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.838 on 700 degrees of freedom
## Multiple R-squared:  0.1286, Adjusted R-squared:  0.07634
## F-statistic: 2.46 on 42 and 700 DF, p-value: 1.632e-06
```

```
#residual analysis plots for subpop 2
plot(model_subpop1_2, which = 1)
```



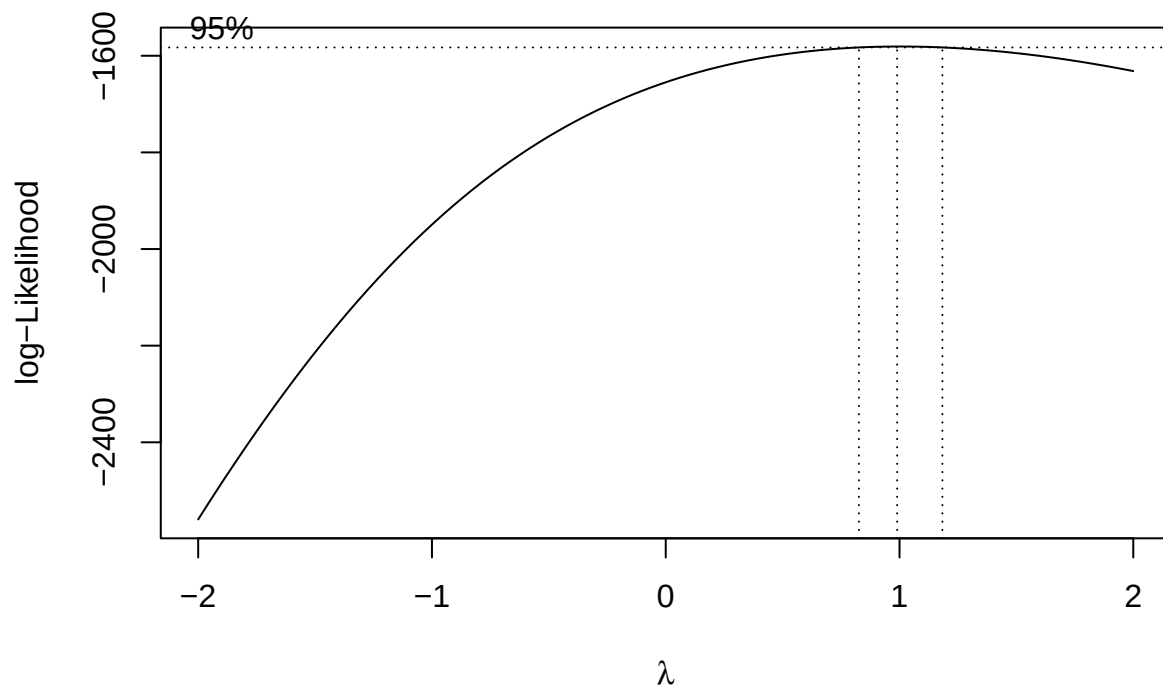
```
lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...
```

```
plot(model_subpop1_2, which = 2)
```

lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...

```
boxcox(model_subpop1_2, plotit=T)
```



```
step(model_subpop1_2, direction="backward")
```

```
## Start: AIC=946.08
## rating ~ age + gender + race + age * genre + age * energy + age *
## danceability + age * happiness + age * bpm + age * loudness_db +
## age * length_sec + race * genre + race * energy + race *
## danceability + race * happiness + race * bpm + race * loudness_db +
```

```

##      race * length_sec + +gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:genre      4      9.737 2374.1 941.14
## - gender:genre    3     12.337 2376.7 943.95
## - age:loudness_db 1      0.051 2364.4 944.10
## - gender:energy   1      0.075 2364.5 944.11
## - age:bpm         1      0.128 2364.5 944.12
## - age:length_sec  1      0.150 2364.5 944.13
## - race:loudness_db 1      0.332 2364.7 944.19
## - age:happiness   1      0.503 2364.9 944.24
## - race:danceability 1      0.616 2365.0 944.28
## - gender:bpm      1      0.885 2365.3 944.36
## - race:happiness  1      0.920 2365.3 944.37
## - gender:loudness_db 1     1.660 2366.1 944.60
## - age:energy      1      3.112 2367.5 945.06
## - race:energy     1      4.791 2369.2 945.59
## - race:length_sec 1      6.350 2370.7 946.08
## <none>                2364.4 946.08
## - gender:length_sec 1      6.605 2371.0 946.15
## - gender:happiness 1      7.593 2372.0 946.46
## - race:bpm        1      8.740 2373.1 946.82
## - age:danceability 1     10.982 2375.4 947.53
## - gender:danceability 1    24.026 2388.4 951.59
## - race:genre      4     45.833 2410.2 952.35
##
## Step:  AIC=941.14
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:happiness + age:bpm + age:loudness_db +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:happiness + race:bpm + race:loudness_db + race:length_sec +
##      gender:genre + gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:loudness_db 1      0.021 2374.2 939.14
## - age:bpm         1      0.057 2374.2 939.15
## - gender:energy   1      0.128 2374.3 939.18
## - age:happiness   1      0.200 2374.3 939.20
## - race:loudness_db 1      0.249 2374.4 939.21
## - race:danceability 1      0.329 2374.5 939.24
## - gender:bpm      1      0.611 2374.7 939.33
## - race:happiness  1      1.465 2375.6 939.59
## - age:length_sec  1      1.601 2375.7 939.64
## - gender:genre    3     14.913 2389.0 939.79
## - gender:loudness_db 1      2.129 2376.3 939.80
## - age:energy      1      3.523 2377.7 940.24
## - race:energy     1      4.452 2378.6 940.53
## - race:length_sec 1      4.984 2379.1 940.69
## <none>                2374.1 941.14
## - gender:length_sec 1      6.499 2380.6 941.17

```

```

## - race:bpm          1      7.803 2381.9 941.57
## - gender:happiness  1      8.175 2382.3 941.69
## - age:danceability  1     12.250 2386.4 942.96
## - race:genre        4     42.193 2416.3 946.22
## - gender:danceability 1     26.426 2400.6 947.36
##
## Step: AIC=939.14
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:length_sec +
## race:genre + race:energy + race:danceability + race:happiness +
## race:bpm + race:loudness_db + race:length_sec + gender:genre +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - age:bpm          1      0.053 2374.2 937.16
## - gender:energy     1      0.135 2374.3 937.18
## - age:happiness     1      0.180 2374.3 937.20
## - race:loudness_db  1      0.261 2374.4 937.22
## - race:danceability 1      0.323 2374.5 937.24
## - gender:bpm        1      0.605 2374.8 937.33
## - race:happiness    1      1.498 2375.7 937.61
## - age:length_sec    1      1.729 2375.9 937.68
## - gender:genre      3     14.954 2389.1 937.81
## - gender:loudness_db 1      2.151 2376.3 937.82
## - race:energy       1      4.632 2378.8 938.59
## - race:length_sec   1      4.963 2379.1 938.69
## <none>                2374.2 939.14
## - gender:length_sec  1      6.631 2380.8 939.21
## - race:bpm          1      7.817 2382.0 939.58
## - gender:happiness  1      8.270 2382.4 939.73
## - age:energy        1      8.827 2383.0 939.90
## - age:danceability  1     12.308 2386.5 940.98
## - race:genre        4     42.194 2416.3 944.23
## - gender:danceability 1     26.486 2400.6 945.39
##
## Step: AIC=937.16
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:genre + gender:energy +
## gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
## gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - gender:energy     1      0.132 2374.3 935.20
## - age:happiness     1      0.155 2374.4 935.21
## - race:loudness_db  1      0.260 2374.5 935.24
## - race:danceability 1      0.315 2374.5 935.26
## - gender:bpm        1      0.627 2374.8 935.36
## - race:happiness    1      1.473 2375.7 935.62
## - age:length_sec    1      1.772 2376.0 935.71

```

```

## - gender:genre          3      14.912 2389.1 935.81
## - gender:loudness_db    1       2.146 2376.3 935.83
## - race:energy           1       4.643 2378.8 936.61
## - race:length_sec       1       4.921 2379.1 936.70
## <none>                  2374.2 937.16
## - gender:length_sec     1       6.624 2380.8 937.23
## - race:bpm              1       7.867 2382.1 937.62
## - gender:happiness      1       8.235 2382.4 937.73
## - age:energy            1       9.374 2383.6 938.09
## - age:danceability      1      12.274 2386.5 938.99
## - race:genre            4      42.467 2416.7 942.33
## - gender:danceability   1      26.620 2400.8 943.44
##
## Step:  AIC=935.2
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:energy +
##         age:danceability + age:happiness + age:length_sec + race:genre +
##         race:energy + race:danceability + race:happiness + race:bpm +
##         race:loudness_db + race:length_sec + gender:genre + gender:danceability +
##         gender:happiness + gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:happiness      1      0.164 2374.5 933.25
## - race:loudness_db    1      0.242 2374.6 933.28
## - race:danceability   1      0.323 2374.7 933.30
## - gender:bpm          1      0.774 2375.1 933.44
## - race:happiness      1      1.422 2375.8 933.65
## - age:length_sec      1      1.706 2376.0 933.73
## - gender:genre        3     15.433 2389.8 934.01
## - gender:loudness_db   1      3.098 2377.4 934.17
## - race:energy         1      4.541 2378.9 934.62
## - race:length_sec     1      4.903 2379.2 934.73
## <none>                2374.3 935.20
## - gender:length_sec   1      6.578 2380.9 935.26
## - race:bpm            1      7.793 2382.1 935.64
## - gender:happiness    1      9.160 2383.5 936.06
## - age:energy          1      9.367 2383.7 936.13
## - age:danceability    1     12.336 2386.7 937.05
## - race:genre          4     42.590 2416.9 940.41
## - gender:danceability  1     26.944 2401.3 941.58
##
## Step:  AIC=933.25
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:energy +
##         age:danceability + age:length_sec + race:genre + race:energy +
##         race:danceability + race:happiness + race:bpm + race:loudness_db +
##         race:length_sec + gender:genre + gender:danceability + gender:happiness +
##         gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:loudness_db    1      0.253 2374.8 931.33
## - race:danceability   1      0.332 2374.8 931.36
## - gender:bpm          1      0.784 2375.3 931.50
## - race:happiness      1      1.504 2376.0 931.72

```

```

## - gender:genre      3      15.570 2390.1 932.11
## - age:length_sec    1       2.998 2377.5 932.19
## - gender:loudness_db 1       3.167 2377.7 932.24
## - race:energy        1       4.651 2379.2 932.71
## - race:length_sec    1       4.857 2379.4 932.77
## <none>                2374.5 933.25
## - gender:length_sec  1       6.503 2381.0 933.28
## - race:bpm           1       7.763 2382.3 933.68
## - gender:happiness   1       9.027 2383.5 934.07
## - age:energy         1      13.781 2388.3 935.55
## - age:danceability   1      16.247 2390.8 936.32
## - race:genre         4      42.449 2416.9 938.42
## - gender:danceability 1      26.849 2401.3 939.61
##
## Step:  AIC=931.33
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:genre + gender:danceability + gender:happiness + gender:bpm +
##      gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability  1      0.367 2375.1 929.45
## - gender:bpm         1      0.852 2375.6 929.60
## - race:happiness     1      1.293 2376.1 929.74
## - gender:genre       3     15.398 2390.2 930.13
## - age:length_sec     1      2.926 2377.7 930.25
## - gender:loudness_db  1      3.043 2377.8 930.28
## - race:length_sec    1      5.183 2379.9 930.95
## - race:energy        1      6.185 2380.9 931.26
## <none>                2374.8 931.33
## - gender:length_sec  1      6.674 2381.4 931.42
## - race:bpm           1      7.646 2382.4 931.72
## - gender:happiness   1      8.929 2383.7 932.12
## - age:energy         1     13.595 2388.3 933.57
## - age:danceability   1     16.233 2391.0 934.39
## - race:genre         4     42.484 2417.2 936.51
## - gender:danceability 1     26.598 2401.3 937.61
##
## Step:  AIC=929.45
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:length_sec + race:genre + race:energy +
##      race:happiness + race:bpm + race:length_sec + gender:genre +
##      gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:bpm         1      0.875 2376.0 927.72
## - race:happiness     1      2.605 2377.7 928.26
## - gender:genre       3     15.532 2390.7 928.29
## - age:length_sec     1      2.821 2377.9 928.33
## - gender:loudness_db  1      3.274 2378.4 928.47

```

```

## - race:length_sec      1      4.816 2379.9 928.95
## - race:energy          1      6.153 2381.3 929.37
## <none>                  2375.1 929.45
## - gender:length_sec    1      7.096 2382.2 929.66
## - race:bpm             1      7.719 2382.8 929.86
## - gender:happiness     1      9.111 2384.2 930.29
## - age:energy           1     13.917 2389.0 931.79
## - age:danceability     1     16.246 2391.4 932.51
## - race:genre           4     44.961 2420.1 935.38
## - gender:danceability  1     28.531 2403.7 936.32
##
## Step:  AIC=927.72
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:length_sec + race:genre + race:energy +
##      race:happiness + race:bpm + race:length_sec + gender:genre +
##      gender:danceability + gender:happiness + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      2.450 2378.4 926.49
## - age:length_sec      1      2.800 2378.8 926.59
## - gender:genre        3     16.078 2392.1 926.73
## - gender:loudness_db  1      3.308 2379.3 926.75
## - race:length_sec     1      4.822 2380.8 927.23
## - race:energy         1      6.119 2382.1 927.63
## <none>                  2376.0 927.72
## - gender:length_sec   1      7.038 2383.0 927.92
## - race:bpm            1      8.182 2384.2 928.27
## - gender:happiness    1      8.633 2384.6 928.41
## - age:energy          1     14.175 2390.2 930.14
## - age:danceability    1     15.991 2392.0 930.70
## - race:genre          4     44.767 2420.8 933.59
## - gender:danceability  1     28.243 2404.2 934.50
##
## Step:  AIC=926.49
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:length_sec + race:genre + race:energy +
##      race:bpm + race:length_sec + gender:genre + gender:danceability +
##      gender:happiness + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:length_sec      1      2.774 2381.2 925.35
## - gender:genre        3     15.956 2394.4 925.45
## - gender:loudness_db  1      3.658 2382.1 925.63
## - race:energy         1      3.906 2382.3 925.70
## <none>                  2378.4 926.49
## - gender:length_sec   1      7.221 2385.7 926.74
## - race:length_sec     1      7.772 2386.2 926.91
## - gender:happiness    1      8.027 2386.5 926.99
## - race:bpm            1      8.241 2386.7 927.05
## - age:energy          1     13.844 2392.3 928.80
## - age:danceability    1     15.193 2393.6 929.22

```

```

## - race:genre          4      46.941 2425.4 933.01
## - gender:danceability 1      28.928 2407.4 933.47
##
## Step: AIC=925.35
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + race:genre + race:energy + race:bpm +
##      race:length_sec + gender:genre + gender:danceability + gender:happiness +
##      gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db 1      3.541 2384.8 924.46
## - gender:genre       3     16.430 2397.7 924.46
## - race:energy        1      3.743 2385.0 924.52
## <none>                2381.2 925.35
## - gender:length_sec  1      6.705 2387.9 925.44
## - race:bpm           1      7.288 2388.5 925.62
## - gender:happiness   1      7.714 2388.9 925.75
## - race:length_sec    1      8.234 2389.4 925.92
## - age:energy         1     13.100 2394.3 927.43
## - age:danceability   1     20.626 2401.8 929.76
## - race:genre         4     46.560 2427.8 931.74
## - gender:danceability 1     28.985 2410.2 932.34
##
## Step: AIC=924.46
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + race:genre + race:energy + race:bpm +
##      race:length_sec + gender:genre + gender:danceability + gender:happiness +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:energy        1      4.446 2389.2 923.84
## - gender:genre       3     17.757 2402.5 923.97
## <none>                2384.8 924.46
## - race:length_sec    1      7.076 2391.8 924.66
## - race:bpm           1      7.086 2391.8 924.66
## - gender:length_sec   1      7.346 2392.1 924.74
## - gender:happiness   1      9.946 2394.7 925.55
## - age:energy         1     13.012 2397.8 926.50
## - loudness_db        1     16.375 2401.1 927.54
## - age:danceability   1     20.842 2405.6 928.92
## - race:genre         4     46.867 2431.6 930.92
## - gender:danceability 1     29.449 2414.2 931.57
##
## Step: AIC=923.84
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + race:genre + race:bpm + race:length_sec +
##      gender:genre + gender:danceability + gender:happiness + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:bpm           1      5.278 2394.5 923.48
## - gender:genre       3     18.483 2407.7 923.56

```

```

## <none>                2389.2 923.84
## - gender:length_sec    1      6.739 2395.9 923.93
## - race:length_sec      1      7.809 2397.0 924.26
## - gender:happiness     1     10.518 2399.7 925.10
## - age:energy           1     12.268 2401.5 925.64
## - loudness_db          1     16.181 2405.4 926.85
## - age:danceability     1     21.781 2411.0 928.58
## - race:genre           4     46.142 2435.3 930.05
## - gender:danceability  1     30.222 2419.4 931.18
##
## Step: AIC=923.48
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + race:genre + race:length_sec + gender:genre +
##      gender:danceability + gender:happiness + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - bpm          1      0.523 2395.0 921.64
## - gender:genre  3     17.826 2412.3 922.99
## <none>                2394.5 923.48
## - gender:length_sec    1      6.753 2401.2 923.57
## - race:length_sec      1      8.277 2402.8 924.04
## - gender:happiness     1     11.019 2405.5 924.89
## - age:energy           1     11.537 2406.0 925.05
## - loudness_db          1     16.289 2410.8 926.52
## - age:danceability     1     21.770 2416.2 928.20
## - race:genre           4     43.852 2438.3 928.96
## - gender:danceability  1     30.661 2425.1 930.93
##
## Step: AIC=921.64
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:energy + age:danceability +
##      race:genre + race:length_sec + gender:genre + gender:danceability +
##      gender:happiness + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - gender:genre  3     17.930 2412.9 921.18
## <none>                2395.0 921.64
## - gender:length_sec    1      6.787 2401.8 921.74
## - race:length_sec      1      8.522 2403.5 922.28
## - gender:happiness     1     10.882 2405.9 923.01
## - age:energy           1     11.281 2406.3 923.13
## - loudness_db          1     16.055 2411.1 924.61
## - age:danceability     1     21.604 2416.6 926.31
## - race:genre           4     44.344 2439.3 927.27
## - gender:danceability  1     30.488 2425.5 929.04
##
## Step: AIC=921.18
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:energy + age:danceability +
##      race:genre + race:length_sec + gender:danceability + gender:happiness +
##      gender:length_sec
##
##              Df Sum of Sq   RSS   AIC

```



```

## <none>                2412.9 921.18
## - race:length_sec      1      7.560 2420.5 921.51
## - age:energy           1     12.072 2425.0 922.89
## - gender:happiness     1     13.906 2426.8 923.45
## - gender:length_sec    1     14.101 2427.0 923.51
## - loudness_db          1     17.532 2430.5 924.56
## - race:genre           4     38.622 2451.6 924.98
## - age:danceability     1     21.255 2434.2 925.70
## - gender:danceability  1     24.641 2437.6 926.73

##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + loudness_db + length_sec + age:energy +
##     age:danceability + race:genre + race:length_sec + gender:danceability +
##     gender:happiness + gender:length_sec, data = first_subpop)
##
## Coefficients:
##              (Intercept)                  age
##              11.3072125                  -0.0803936
##              genderNot Male                raceWhite
##              0.3815542                  -2.7522149
##              genreJazz                    genreOther
##              -0.7124181                  -0.3804099
##              genrePop                    genreRock
##              -1.3932352                  -1.2816427
##              energy                      danceability
##              0.0208016                  -0.0751822
##              happiness                  loudness_db
##              0.0118214                  0.0693646
##              length_sec                 age:energy
##              0.0006032                  -0.0010484
##              age:danceability            raceWhite:genreJazz
##              0.0018960                  1.6225519
##              raceWhite:genreOther        raceWhite:genrePop
##              0.4957498                  1.3548092
##              raceWhite:genreRock         raceWhite:length_sec
##              1.9127645                  0.0044321
## genderNot Male:danceability            genderNot Male:happiness
##              0.0324124                  -0.0148272
## genderNot Male:length_sec
##              -0.0063696

new_model_subpop1_2 <- lm(data = first_subpop, rating ~ age + gender + race + genre + energy + danceability +
summary(new_model_subpop1_2)

##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + loudness_db + length_sec + age *
##     energy + age * danceability + race * genre + race * length_sec +
##     gender * genre + gender * energy + gender * danceability +
##     gender * happiness + gender * length_sec, data = first_subpop)
##
## Residuals:

```

```

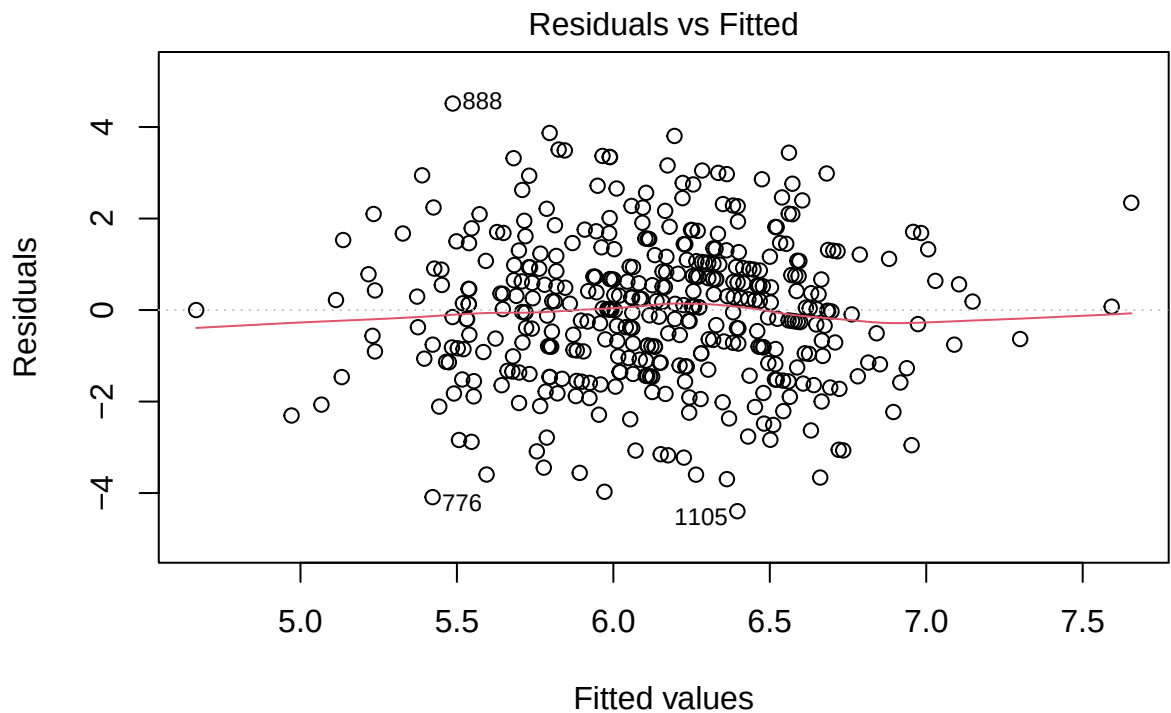
##      Min      1Q  Median      3Q      Max
## -4.790 -1.204 -0.055  1.241  4.665
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    11.5280368   1.9853914   5.806 9.59e-09 ***
## age            -0.0802383   0.0450324  -1.782 0.075207 .
## genderNot Male    0.9740254   1.1591996    0.840 0.401045
## raceWhite       -3.3083089   1.5055817  -2.197 0.028314 *
## genreJazz       -0.8264962   1.6879220  -0.490 0.624529
## genreOther      -0.2822773   1.3238375  -0.213 0.831211
## genrePop        -1.5475137   1.3238625  -1.169 0.242817
## genreRock       -1.3232179   1.3462558  -0.983 0.325995
## energy           0.0224552   0.0165727    1.355 0.175860
## danceability    -0.0763462   0.0224371  -3.403 0.000704 ***
## happiness        0.0103878   0.0047353    2.194 0.028578 *
## loudness_db      0.0681777   0.0305130    2.234 0.025766 *
## length_sec      -0.0001428   0.0028391  -0.050 0.959909
## age:energy       -0.0010267   0.0005527  -1.857 0.063654 .
## age:danceability  0.0019020   0.0007538    2.523 0.011844 *
## raceWhite:genreJazz  2.1426357   2.0393118    1.051 0.293766
## raceWhite:genreOther  0.9029494   1.4119739    0.639 0.522706
## raceWhite:genrePop   1.8528869   1.3995658    1.324 0.185958
## raceWhite:genreRock   2.3812477   1.4186333    1.679 0.093676 .
## raceWhite:length_sec  0.0048314   0.0029624    1.631 0.103347
## genderNot Male:genreJazz      NA         NA         NA         NA
## genderNot Male:genreOther -1.6538896   0.8904267  -1.857 0.063663 .
## genderNot Male:genrePop  -1.0767216   0.8414050  -1.280 0.201075
## genderNot Male:genreRock -1.4211838   0.8901484  -1.597 0.110803
## genderNot Male:energy    -0.0065133   0.0091288  -0.713 0.475774
## genderNot Male:danceability  0.0372524   0.0122757    3.035 0.002495 **
## genderNot Male:happiness  -0.0107106   0.0081561  -1.313 0.189534
## genderNot Male:length_sec -0.0040748   0.0032808  -1.242 0.214639
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.828 on 716 degrees of freedom
## Multiple R-squared:  0.118, Adjusted R-squared:  0.08594
## F-statistic: 3.683 on 26 and 716 DF, p-value: 3.351e-09
model_subpop2_1 <- lm(data = second_subpop, rating ~ age + race + gender + genre + energy + danceability +
summary(model_subpop2_1)

##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##     danceability + happiness + bpm + loudness_db + length_sec,
##     data = second_subpop)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -4.3964 -1.0558  0.0392  0.9655  4.5133
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)

```

```
## (Intercept)      5.016752    1.960993    2.558  0.01089 *
## age             -0.001924    0.012044   -0.160  0.87313
## raceWhite       -0.128474    0.187366   -0.686  0.49331
## genderNot Male   0.484635    0.162164    2.989  0.00298 **
## genreOther       0.707159    1.615313    0.438  0.66178
## genrePop         0.748135    1.619543    0.462  0.64438
## genreRock        0.657571    1.637667    0.402  0.68825
## energy          -0.015693    0.007433   -2.111  0.03537 *
## danceability     0.003811    0.007058    0.540  0.58957
## happiness        0.011840    0.004400    2.691  0.00742 **
## bpm             -0.001863    0.003034   -0.614  0.53954
## loudness_db      0.081621    0.050192    1.626  0.10470
## length_sec       0.004682    0.001934    2.420  0.01595 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.595 on 398 degrees of freedom
## Multiple R-squared:  0.07127,    Adjusted R-squared:  0.04327
## F-statistic: 2.545 on 12 and 398 DF,  p-value: 0.002998
```

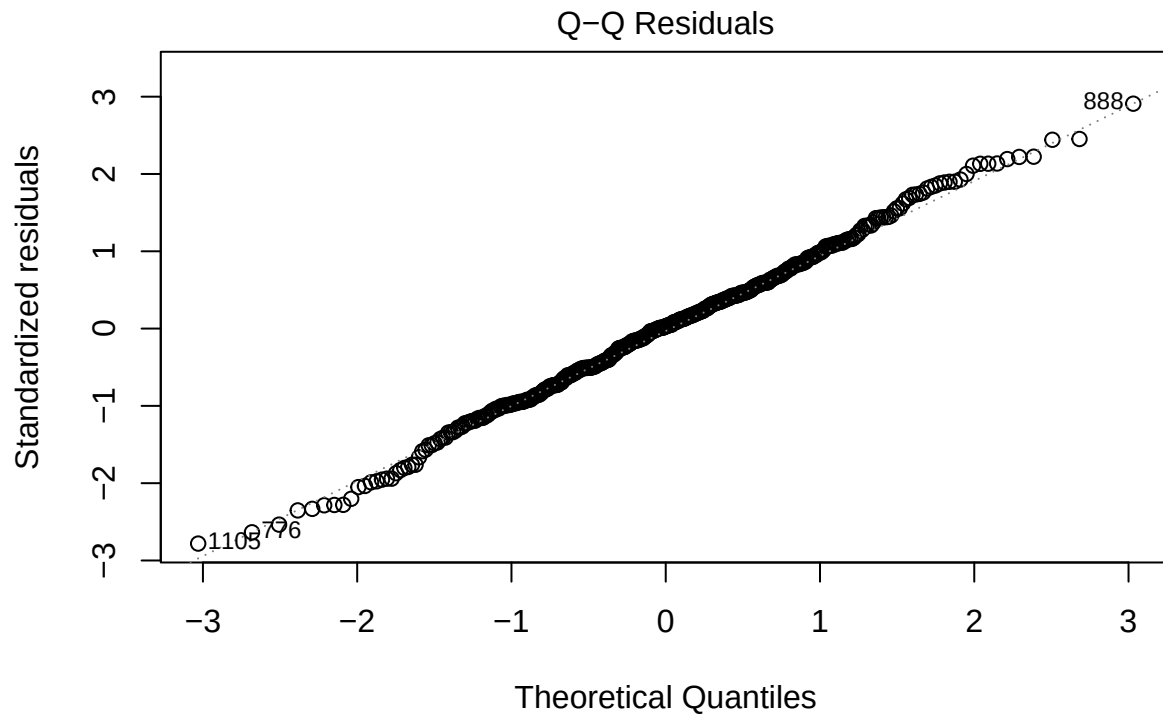
```
plot(model_subpop2_1, which = 1)
```



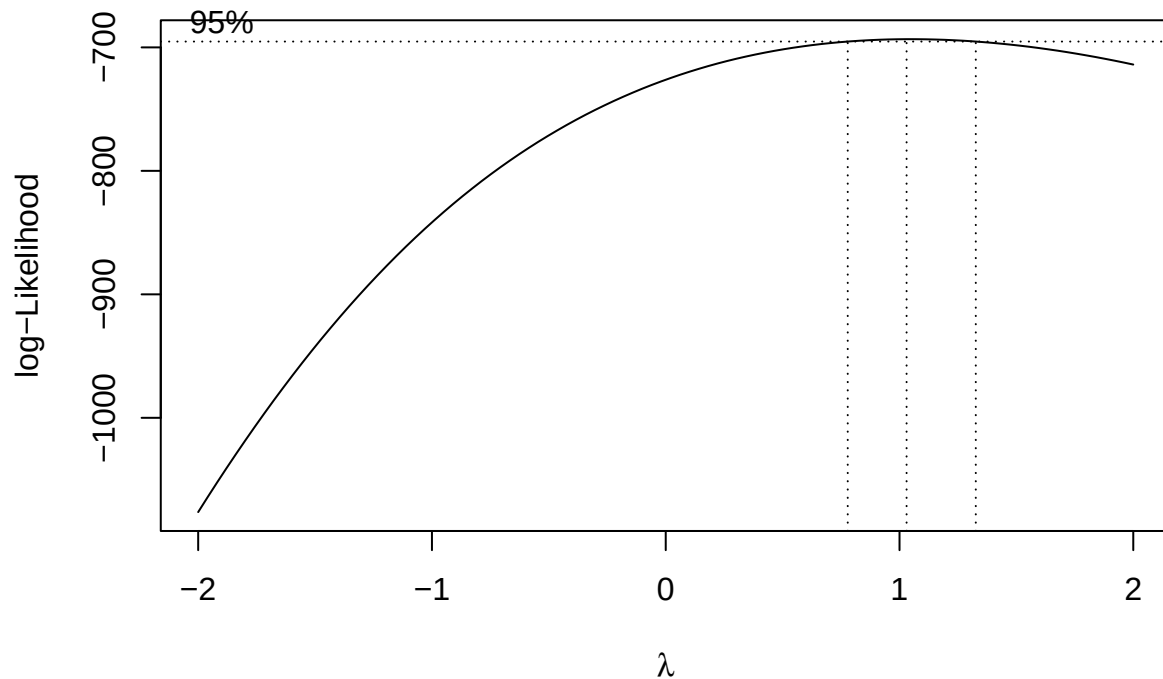
lm(rating ~ age + race + gender + genre + energy + danceability + happiness ...

```
plot(model_subpop2_1, which = 2)
```

```
## Warning: not plotting observations with leverage one:
##      89
```



```
boxcox(model_subpop2_1, plotit=T)
```



```
model_subpop2_2 <- lm(data = second_subpop, rating ~ age + gender + race + age*genre + age*energy + age*  
summary(model_subpop2_2)
```

```
##  
## Call:  
## lm(formula = rating ~ age + gender + race + age * genre + age *  
##     energy + age * danceability + age * happiness + age * bpm +
```

```

##      age * loudness_db + age * length_sec + race * genre + race *
##      energy + race * danceability + race * happiness + race *
##      bpm + race * loudness_db + race * length_sec + +gender *
##      genre + gender * energy + gender * danceability + gender *
##      happiness + gender * bpm + gender * loudness_db + gender *
##      length_sec, data = second_subpop)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -4.2473 -1.0138  0.0664  0.9589  3.6915
##
## Coefficients: (3 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.962e+00  4.880e+00  -0.402  0.68790
## age              2.331e-01  1.437e-01   1.622  0.10573
## genderNot Male  -9.228e-01  2.197e+00  -0.420  0.67474
## raceWhite       -1.117e+00  2.418e+00  -0.462  0.64441
## genreOther       4.728e+00  2.903e+00   1.628  0.10428
## genrePop         6.778e+00  2.969e+00   2.283  0.02300 *
## genreRock        1.609e+00  1.681e+00   0.957  0.33913
## energy           1.072e-02  2.993e-02   0.358  0.72044
## danceability     1.117e-02  3.239e-02   0.345  0.73033
## happiness        -6.277e-03  1.940e-02  -0.324  0.74642
## bpm              -1.537e-02  1.478e-02  -1.040  0.29899
## loudness_db      -8.527e-03  2.325e-01  -0.037  0.97076
## length_sec       1.609e-02  9.603e-03   1.676  0.09460 .
## age:genreOther   -9.014e-02  5.513e-02  -1.635  0.10289
## age:genrePop     -1.817e-01  5.882e-02  -3.089  0.00216 **
## age:genreRock    NA          NA          NA      NA
## age:energy       -1.092e-03  9.919e-04  -1.100  0.27183
## age:danceability  4.116e-05  1.026e-03   0.040  0.96802
## age:happiness     6.305e-04  6.349e-04   0.993  0.32131
## age:bpm           3.916e-04  4.937e-04   0.793  0.42812
## age:loudness_db  4.001e-03  7.673e-03   0.521  0.60238
## age:length_sec   -5.426e-04  3.175e-04  -1.709  0.08824 .
## raceWhite:genreOther -6.881e-01  1.096e+00  -0.628  0.53061
## raceWhite:genrePop  6.208e-02  1.107e+00   0.056  0.95533
## raceWhite:genreRock NA          NA          NA      NA
## raceWhite:energy  -2.158e-02  1.578e-02  -1.368  0.17215
## raceWhite:danceability 1.540e-02  1.591e-02   0.968  0.33351
## raceWhite:happiness -6.471e-03  1.026e-02  -0.631  0.52846
## raceWhite:bpm     1.194e-02  6.783e-03   1.760  0.07915 .
## raceWhite:loudness_db 1.438e-01  1.048e-01   1.372  0.17094
## raceWhite:length_sec 7.067e-03  4.177e-03   1.692  0.09149 .
## genderNot Male:genreOther -1.591e-01  8.633e-01  -0.184  0.85384
## genderNot Male:genrePop  1.766e-01  9.150e-01   0.193  0.84703
## genderNot Male:genreRock NA          NA          NA      NA
## genderNot Male:energy  2.686e-02  1.522e-02   1.765  0.07845 .
## genderNot Male:danceability -2.706e-02  1.442e-02  -1.877  0.06128 .
## genderNot Male:happiness 2.947e-03  9.101e-03   0.324  0.74625
## genderNot Male:bpm    -1.735e-04  6.152e-03  -0.028  0.97752
## genderNot Male:loudness_db -1.377e-01  1.032e-01  -1.334  0.18311
## genderNot Male:length_sec 2.612e-03  4.072e-03   0.641  0.52162
## ---

```

```

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.568 on 374 degrees of freedom
## Multiple R-squared:  0.1563, Adjusted R-squared:  0.0751
## F-statistic: 1.925 on 36 and 374 DF,  p-value: 0.001511
step(model_subpop2_2, direction="backward")

## Start:  AIC=405.11
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
##
##           Df Sum of Sq  RSS   AIC
## - gender:genre      2    1.4843 921.34 401.78
## - gender:bpm        1     0.0020 919.86 403.11
## - age:danceability   1     0.0040 919.86 403.11
## - gender:happiness   1     0.2579 920.12 403.23
## - age:loudness_db    1     0.6687 920.53 403.41
## - race:happiness     1     0.9791 920.84 403.55
## - gender:length_sec  1     1.0120 920.87 403.57
## - age:bpm           1     1.5477 921.41 403.80
## - race:danceability  1     2.3061 922.17 404.14
## - age:happiness      1     2.4257 922.29 404.20
## - race:genre         2     7.1470 927.01 404.29
## - age:energy         1     2.9786 922.84 404.44
## - gender:loudness_db  1     4.3750 924.23 405.06
## <none>                                919.86 405.11
## - race:energy        1     4.6023 924.46 405.16
## - race:loudness_db   1     4.6285 924.49 405.18
## - race:length_sec    1     7.0406 926.90 406.25
## - age:length_sec     1     7.1854 927.05 406.31
## - race:bpm          1     7.6224 927.48 406.50
## - gender:energy      1     7.6581 927.52 406.52
## - gender:danceability 1     8.6664 928.53 406.97
## - age:genre          2    28.9260 948.79 413.84
##
## Step:  AIC=401.78
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:genre +
##   age:energy + age:danceability + age:happiness + age:bpm +
##   age:loudness_db + age:length_sec + race:genre + race:energy +
##   race:danceability + race:happiness + race:bpm + race:loudness_db +
##   race:length_sec + gender:energy + gender:danceability + gender:happiness +
##   gender:bpm + gender:loudness_db + gender:length_sec
##
##
##           Df Sum of Sq  RSS   AIC
## - gender:bpm        1     0.0096 921.35 399.78
## - age:danceability   1     0.0135 921.36 399.78
## - gender:happiness   1     0.3309 921.68 399.92
## - age:loudness_db    1     0.5635 921.91 400.03

```

```

## - race:happiness      1    0.9249 922.27 400.19
## - gender:length_sec   1    1.0352 922.38 400.24
## - age:bpm             1    1.2750 922.62 400.34
## - race:danceability   1    2.4083 923.75 400.85
## - age:happiness       1    2.5064 923.85 400.89
## - age:energy          1    2.8033 924.15 401.02
## - race:genre          2    8.2973 929.64 401.46
## <none>                921.34 401.78
## - race:energy         1    4.6793 926.02 401.86
## - gender:loudness_db  1    4.8048 926.15 401.91
## - race:loudness_db    1    5.1168 926.46 402.05
## - race:length_sec     1    6.7872 928.13 402.79
## - age:length_sec      1    6.9356 928.28 402.86
## - gender:energy       1    7.3142 928.66 403.03
## - race:bpm            1    7.3955 928.74 403.06
## - gender:danceability 1    10.9444 932.29 404.63
## - age:genre           2    30.9634 952.31 411.36
##
## Step: AIC=399.78
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + age:happiness + age:bpm +
## age:loudness_db + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:loudness_db +
## race:length_sec + gender:energy + gender:danceability + gender:happiness +
## gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq    RSS    AIC
## - age:danceability  1    0.0132 921.37 397.79
## - gender:happiness  1    0.3392 921.69 397.93
## - age:loudness_db   1    0.5612 921.91 398.03
## - race:happiness    1    0.9188 922.27 398.19
## - gender:length_sec 1    1.0482 922.40 398.25
## - age:bpm           1    1.2726 922.63 398.35
## - race:danceability 1    2.4025 923.76 398.85
## - age:happiness     1    2.4997 923.85 398.89
## - age:energy        1    2.7974 924.15 399.03
## - race:genre        2    8.3276 929.68 399.48
## <none>              921.35 399.78
## - race:energy       1    4.7019 926.06 399.87
## - gender:loudness_db 1    4.8105 926.16 399.92
## - race:loudness_db  1    5.1366 926.49 400.07
## - race:length_sec   1    6.7776 928.13 400.79
## - age:length_sec    1    6.9392 928.29 400.86
## - gender:energy     1    7.3801 928.73 401.06
## - race:bpm          1    7.4006 928.75 401.07
## - gender:danceability 1    11.0170 932.37 402.67
## - age:genre         2    31.4020 952.76 409.55
##
## Step: AIC=397.79
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:genre + race:energy + race:danceability +

```

```

##      race:happiness + race:bpm + race:loudness_db + race:length_sec +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:happiness      1      0.342 921.71 395.94
## - age:loudness_db        1      0.582 921.95 396.05
## - race:happiness         1      0.918 922.29 396.20
## - gender:length_sec       1      1.042 922.41 396.25
## - age:bpm                1      1.264 922.63 396.35
## - race:danceability       1      2.390 923.76 396.85
## - age:energy              1      2.786 924.15 397.03
## - age:happiness          1      2.866 924.23 397.06
## - race:genre              2      8.324 929.69 397.48
## <none>                    921.37 397.79
## - race:energy             1      4.694 926.06 397.87
## - gender:loudness_db      1      4.809 926.18 397.93
## - race:loudness_db        1      5.137 926.50 398.07
## - race:length_sec         1      6.857 928.22 398.83
## - age:length_sec          1      7.080 928.45 398.93
## - gender:energy           1      7.379 928.75 399.06
## - race:bpm                1      7.474 928.84 399.11
## - gender:danceability     1     11.127 932.49 400.72
## - age:genre               2     32.202 953.57 407.91
##
## Step:  AIC=395.94
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:loudness_db +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:happiness + race:bpm + race:loudness_db + race:length_sec +
##      gender:energy + gender:danceability + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:loudness_db        1      0.631 922.34 394.22
## - race:happiness         1      0.834 922.54 394.31
## - gender:length_sec       1      0.942 922.65 394.36
## - age:bpm                1      1.397 923.11 394.56
## - race:danceability       1      2.286 924.00 394.96
## - age:happiness          1      2.736 924.44 395.16
## - age:energy              1      2.898 924.61 395.23
## - race:genre              2      8.603 930.31 395.76
## <none>                    921.71 395.94
## - race:energy             1      4.766 926.47 396.06
## - gender:loudness_db      1      5.107 926.82 396.21
## - race:loudness_db        1      5.267 926.98 396.28
## - race:length_sec         1      6.873 928.58 396.99
## - age:length_sec          1      7.196 928.90 397.13
## - race:bpm                1      7.406 929.11 397.23
## - gender:energy           1     10.861 932.57 398.75
## - gender:danceability     1     11.549 933.26 399.06
## - age:genre               2     32.582 954.29 406.22
##

```



```

## Step: AIC=394.22
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:energy + gender:danceability +
## gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - gender:length_sec 1 0.774 923.11 392.56
## - race:happiness 1 0.802 923.14 392.58
## - age:bpm 1 1.482 923.82 392.88
## - race:danceability 1 2.292 924.63 393.24
## - age:energy 1 2.306 924.65 393.25
## - age:happiness 1 2.365 924.70 393.27
## - race:genre 2 8.983 931.32 394.20
## <none> 922.34 394.22
## - race:energy 1 4.973 927.31 394.43
## - gender:loudness_db 1 5.447 927.79 394.64
## - race:loudness_db 1 5.577 927.92 394.70
## - race:length_sec 1 6.939 929.28 395.30
## - age:length_sec 1 7.032 929.37 395.34
## - race:bpm 1 7.616 929.96 395.60
## - gender:energy 1 10.863 933.20 397.03
## - gender:danceability 1 11.212 933.55 397.19
## - age:genre 2 32.096 954.44 404.28
##
## Step: AIC=392.56
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:energy + gender:danceability +
## gender:loudness_db
##
##
## Df Sum of Sq RSS AIC
## - race:happiness 1 0.7424 923.86 390.89
## - age:bpm 1 1.3493 924.46 391.16
## - race:danceability 1 2.2669 925.38 391.57
## - age:energy 1 2.3645 925.48 391.62
## - age:happiness 1 2.5312 925.64 391.69
## - race:genre 2 8.5359 931.65 392.35
## <none> 923.11 392.56
## - race:energy 1 5.1581 928.27 392.85
## - race:loudness_db 1 5.5774 928.69 393.04
## - gender:loudness_db 1 5.7451 928.86 393.11
## - age:length_sec 1 7.1242 930.24 393.72
## - race:bpm 1 7.5345 930.65 393.91
## - race:length_sec 1 8.6645 931.78 394.40
## - gender:energy 1 10.7990 933.91 395.34
## - gender:danceability 1 11.5194 934.63 395.66
## - age:genre 2 31.5306 954.64 402.37
##
## Step: AIC=390.89

```

```

## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:genre +
##   age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##   race:energy + race:danceability + race:bpm + race:loudness_db +
##   race:length_sec + gender:energy + gender:danceability + gender:loudness_db
##
##           Df Sum of Sq   RSS   AIC
## - age:bpm      1    1.5613 925.42 389.59
## - race:danceability 1    1.5730 925.43 389.59
## - age:energy     1    2.2163 926.07 389.88
## - age:happiness  1    2.6159 926.47 390.06
## - race:genre     2    8.0500 931.91 390.46
## <none>                      923.86 390.89
## - gender:loudness_db 1    5.8441 929.70 391.49
## - race:loudness_db  1    6.0102 929.87 391.56
## - age:length_sec    1    6.7965 930.65 391.91
## - race:bpm          1    7.7711 931.63 392.34
## - race:energy       1    8.6649 932.52 392.73
## - race:length_sec   1    9.7308 933.59 393.20
## - gender:energy     1   10.6734 934.53 393.62
## - gender:danceability 1   11.4534 935.31 393.96
## - age:genre        2   31.4595 955.32 400.66
##
## Step:  AIC=389.59
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:genre +
##   age:energy + age:happiness + age:length_sec + race:genre +
##   race:energy + race:danceability + race:bpm + race:loudness_db +
##   race:length_sec + gender:energy + gender:danceability + gender:loudness_db
##
##           Df Sum of Sq   RSS   AIC
## - age:energy     1    1.3646 926.78 388.19
## - race:danceability 1    1.6576 927.07 388.32
## - age:happiness  1    1.8589 927.28 388.41
## - race:genre     2    7.9712 933.39 389.11
## <none>                      925.42 389.59
## - gender:loudness_db 1    5.3774 930.79 389.97
## - age:length_sec    1    5.5786 931.00 390.06
## - race:loudness_db  1    6.3500 931.77 390.40
## - race:bpm          1    7.5795 933.00 390.94
## - race:energy       1    8.7319 934.15 391.45
## - race:length_sec   1    9.5836 935.00 391.82
## - gender:energy     1    9.9288 935.35 391.97
## - gender:danceability 1   11.0635 936.48 392.47
## - age:genre        2   30.0016 955.42 398.70
##
## Step:  AIC=388.19
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:genre +
##   age:happiness + age:length_sec + race:genre + race:energy +
##   race:danceability + race:bpm + race:loudness_db + race:length_sec +
##   gender:energy + gender:danceability + gender:loudness_db
##
##           Df Sum of Sq   RSS   AIC

```

```

## - age:happiness      1      1.0907 927.87 386.68
## - race:danceability  1      1.6670 928.45 386.93
## - race:genre         2      8.6525 935.43 388.01
## <none>                926.78 388.19
## - gender:loudness_db 1      5.1876 931.97 388.49
## - race:loudness_db   1      6.2395 933.02 388.95
## - race:bpm           1      7.2407 934.02 389.39
## - age:length_sec     1      8.5719 935.35 389.98
## - race:energy        1      8.7457 935.53 390.05
## - race:length_sec    1      9.6912 936.47 390.47
## - gender:energy      1      9.9734 936.76 390.59
## - gender:danceability 1     11.2363 938.02 391.15
## - age:genre          2     28.7474 955.53 396.75
##
## Step:  AIC=386.68
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:bpm + race:loudness_db + race:length_sec + gender:energy +
##      gender:danceability + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability  1      1.6792 929.55 385.42
## - race:genre         2      8.2264 936.10 386.31
## <none>                927.87 386.68
## - gender:loudness_db  1      4.7561 932.63 386.78
## - race:loudness_db   1      6.7349 934.61 387.65
## - race:bpm           1      6.9644 934.84 387.75
## - race:length_sec    1      9.3496 937.22 388.80
## - gender:energy      1      9.4701 937.34 388.85
## - race:energy        1      9.5804 937.45 388.90
## - gender:danceability 1     11.4918 939.36 389.74
## - age:length_sec     1     13.4189 941.29 390.58
## - happiness          1     14.8714 942.74 391.21
## - age:genre          2     29.1701 957.04 395.40
##
## Step:  AIC=385.42
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:length_sec + race:genre + race:energy + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - race:genre         2      6.9875 936.54 384.50
## <none>                929.55 385.42
## - gender:loudness_db  1      4.7120 934.26 385.50
## - race:bpm           1      5.8556 935.41 386.00
## - race:loudness_db   1      7.1603 936.71 386.57
## - race:length_sec    1      8.5084 938.06 387.17
## - race:energy        1      8.9631 938.51 387.36
## - gender:energy      1      9.5329 939.08 387.61
## - gender:danceability 1     11.7423 941.29 388.58
## - age:length_sec     1     12.6574 942.21 388.98
## - happiness          1     15.0793 944.63 390.03

```

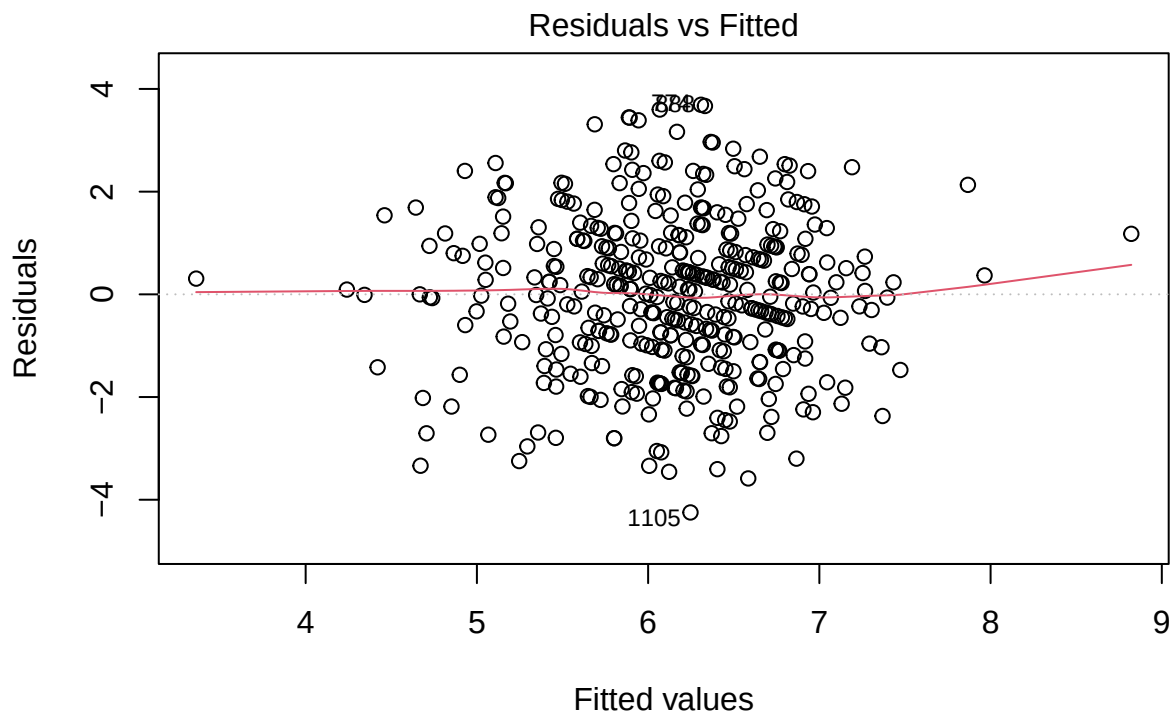
```

## - age:genre          2    28.5969 958.15 393.87
##
## Step: AIC=384.5
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:length_sec + race:energy + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db  1      4.463 941.00 384.45
## <none>                                936.54 384.50
## - race:bpm            1      5.179 941.72 384.77
## - race:loudness_db    1      6.775 943.31 385.46
## - race:length_sec     1      9.438 945.98 386.62
## - gender:energy       1     10.095 946.63 386.91
## - race:energy         1     10.845 947.38 387.23
## - age:length_sec      1     11.039 947.58 387.32
## - gender:danceability 1     11.255 947.79 387.41
## - happiness          1     16.649 953.19 389.74
## - age:genre          2     32.429 968.97 394.49
##
## Step: AIC=384.45
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:length_sec + race:energy + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability
##
##              Df Sum of Sq    RSS    AIC
## <none>                                941.00 384.45
## - race:bpm            1      5.4729 946.47 384.84
## - gender:energy       1      5.6331 946.64 384.91
## - race:loudness_db    1      7.2127 948.21 385.59
## - age:length_sec      1      9.3183 950.32 386.50
## - race:length_sec     1      9.7073 950.71 386.67
## - race:energy         1     10.7980 951.80 387.14
## - gender:danceability 1     12.9272 953.93 388.06
## - happiness          1     17.4717 958.47 390.01
## - age:genre          2     30.8859 971.89 393.73
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##      danceability + happiness + bpm + loudness_db + length_sec +
##      age:genre + age:length_sec + race:energy + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability, data = second_subpop)
##
## Coefficients:
##              (Intercept)              age
##              -1.3452497              0.1796533
##              genderNot Male              raceWhite
##              1.4590885              -0.1107791
##              genreOther              genrePop
##              3.1521627              5.8477239
##              genreRock              energy

```

```
##          1.2441293          -0.0129477
##          danceability          happiness
##          0.0177017          0.0116070
##          bpm          loudness_db
##          -0.0041044          0.0309261
##          length_sec          age:genreOther
##          0.0157296          -0.0621346
##          age:genrePop          age:genreRock
##          -0.1589752          NA
##          age:length_sec          raceWhite:energy
##          -0.0004902          -0.0290771
##          raceWhite:bpm          raceWhite:loudness_db
##          0.0097808          0.1737385
##          raceWhite:length_sec          genderNot Male:energy
##          0.0077875          0.0151987
## genderNot Male:danceability
##          -0.0284454
```

```
plot(model_subpop2_2, which = 1)
```

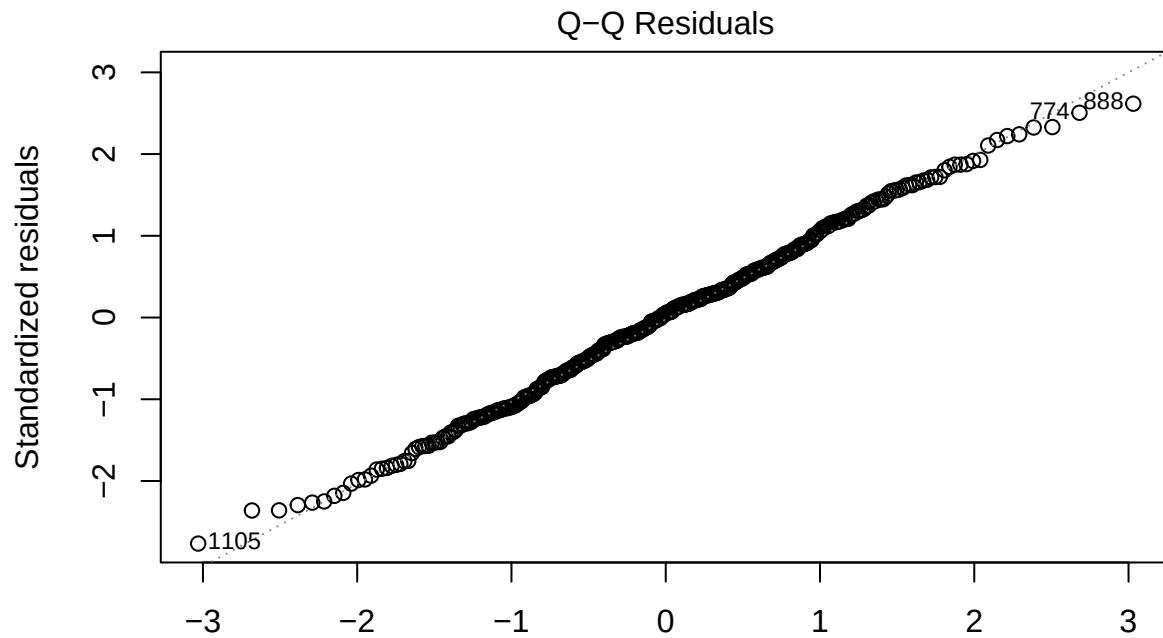


`lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...`

```
plot(model_subpop2_2, which = 2)
```

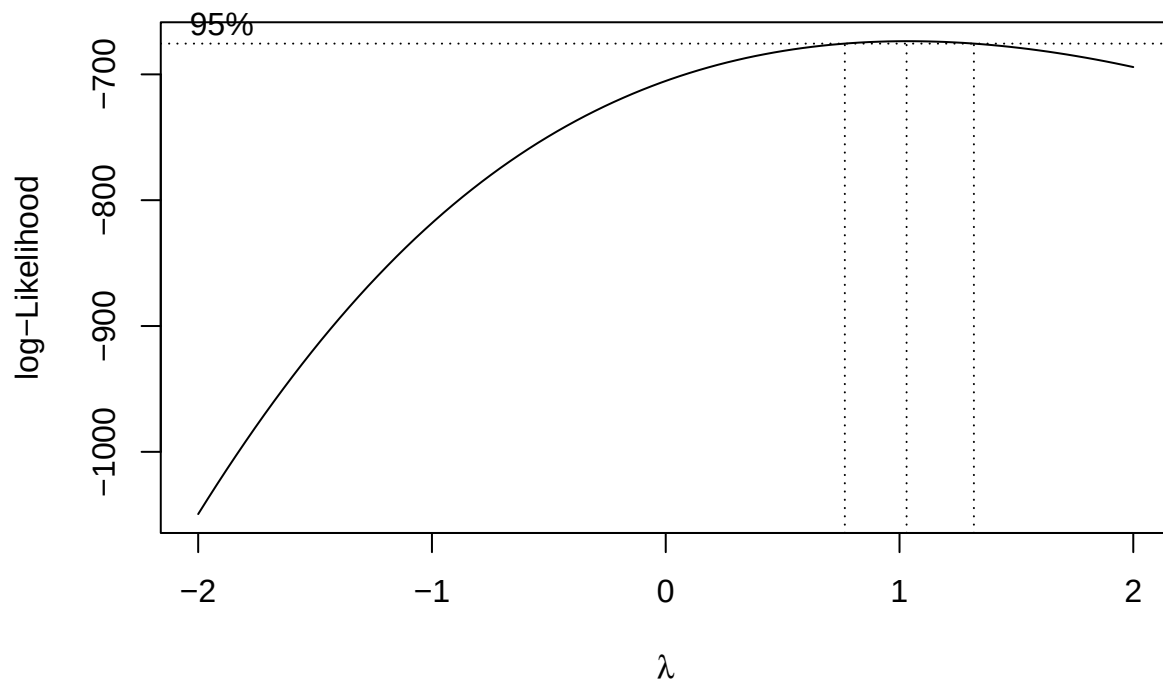
```
## Warning: not plotting observations with leverage one:
```

```
##      89
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...

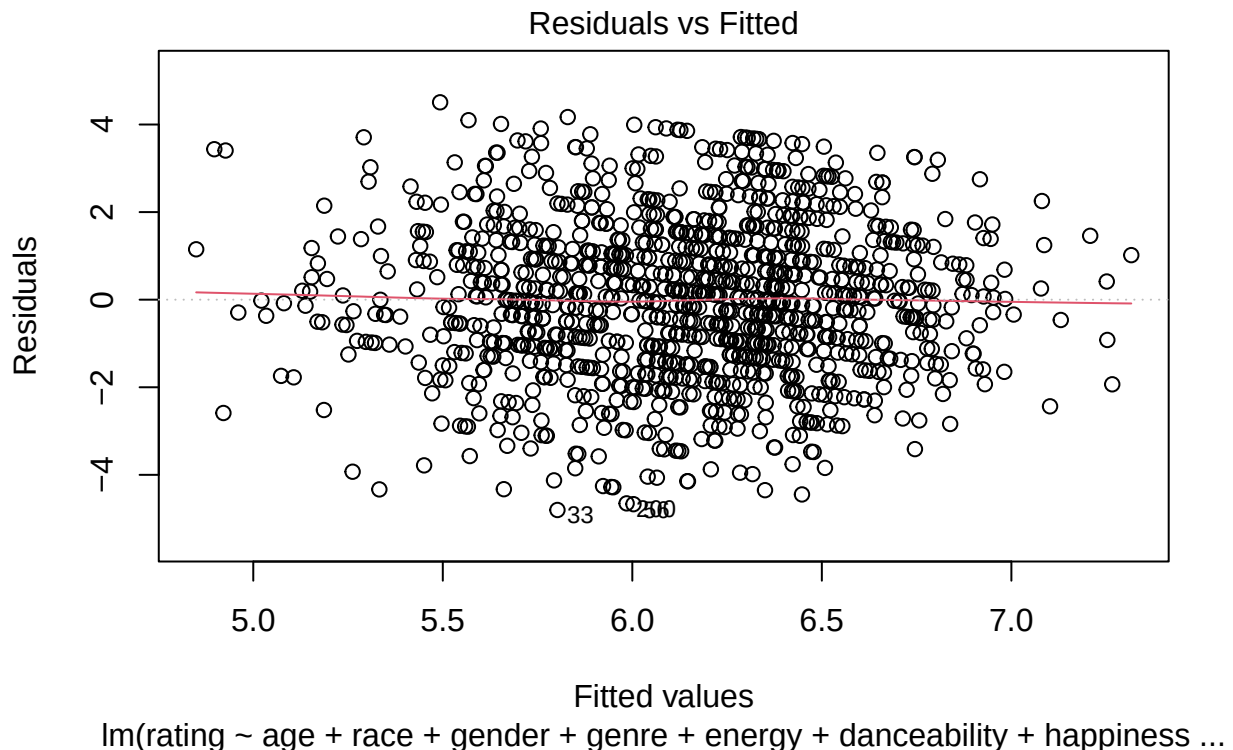
```
boxcox(model_subpop2_2, plotit=T)
```



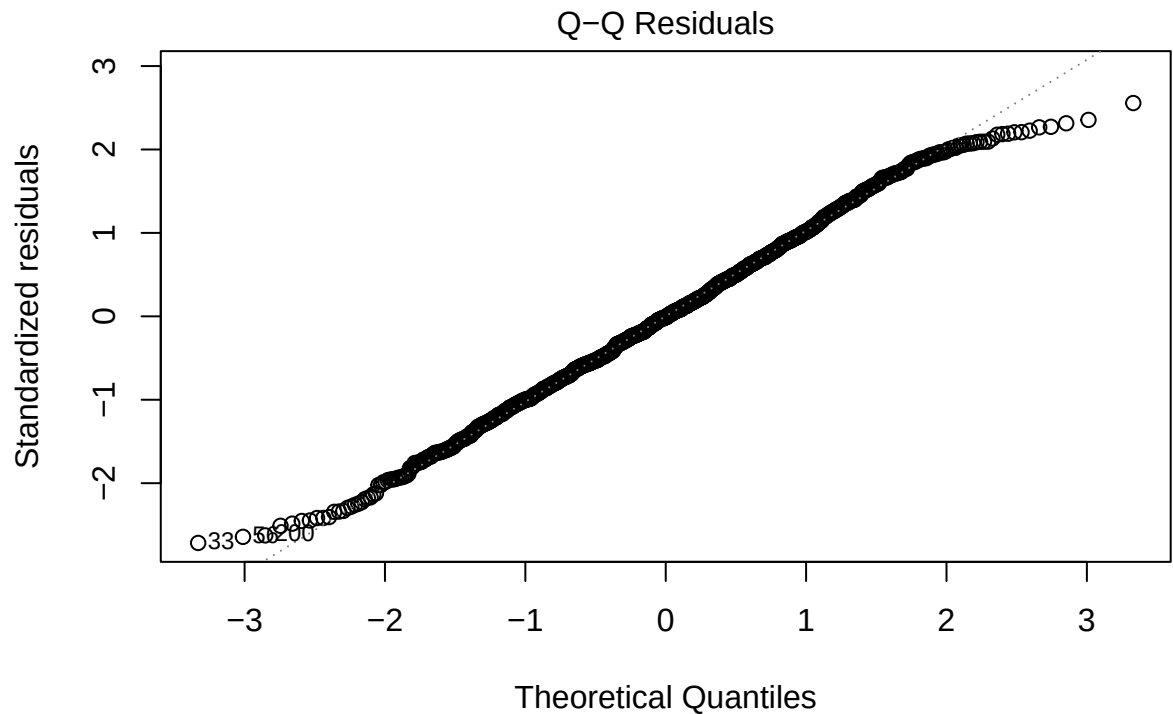
```
model_total_1 <- lm(data = cleaned_data, rating ~ age + race + gender + genre + energy + danceability +  
summary(model_total_1)
```

```
##  
## Call:  
## lm(formula = rating ~ age + race + gender + genre + energy +  
##     danceability + happiness + bpm + loudness_db + length_sec,
```

```
##      data = cleaned_data)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -4.8024 -1.2244 -0.0203  1.2248  4.5071
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.0722615   0.6747344   10.482 < 2e-16 ***
## age          -0.0235027   0.0080080    -2.935  0.003403 **
## raceWhite     -0.4425431   0.1231832    -3.593  0.000341 ***
## genderNot Male  0.2008111   0.1139569     1.762  0.078309 .
## genreJazz      1.1915765   0.8098788     1.471  0.141485
## genreOther     0.2896938   0.3717489     0.779  0.435981
## genrePop      -0.0309389   0.3602833    -0.086  0.931582
## genreRock      0.5806698   0.3666694     1.584  0.113555
## energy        -0.0086781   0.0045002    -1.928  0.054054 .
## danceability  -0.0098823   0.0043489    -2.272  0.023249 *
## happiness      0.0103522   0.0028992     3.571  0.000371 ***
## bpm           -0.0006552   0.0021639    -0.303  0.762097
## loudness_db    0.0442563   0.0223927     1.976  0.048353 *
## length_sec     0.0027224   0.0011510     2.365  0.018188 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.78 on 1140 degrees of freedom
## Multiple R-squared:  0.05086,    Adjusted R-squared:  0.04003
## F-statistic: 4.699 on 13 and 1140 DF,  p-value: 5.732e-08
plot(model_total_1, which = 1)
```

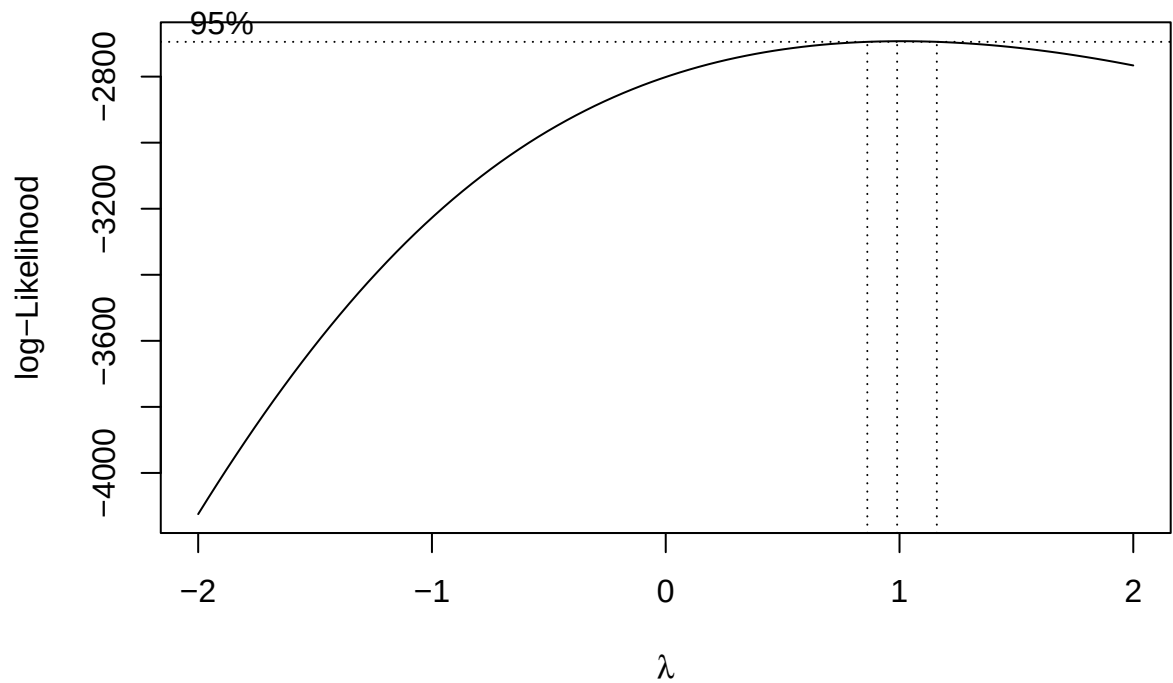


```
plot(model_total_1, which = 2)
```



lm(rating ~ age + race + gender + genre + energy + danceability + happiness ...

```
boxcox(model_total_1, plotit=T)
```



```
model_total_2 <- lm(data = cleaned_data, rating ~ age + gender + race + age*genre + age*energy + age*danceability + age*happiness)
summary(model_total_2)
```

```
##
```



```
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##     energy + age * danceability + age * happiness + age * bpm +
##     age * loudness_db + age * length_sec + race * genre + race *
##     energy + race * danceability + race * happiness + race *
##     bpm + race * loudness_db + race * length_sec + +gender *
##     genre + gender * energy + gender * danceability + gender *
##     happiness + gender * bpm + gender * loudness_db + gender *
##     length_sec, data = cleaned_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7534 -1.1493 -0.0184  1.1834  4.4488
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      9.492e+00  3.315e+00   2.863  0.00427 **
## age              -6.391e-02  9.526e-02  -0.671  0.50243
## genderNot Male    3.706e-01  1.367e+00   0.271  0.78633
## raceWhite        -1.378e+00  1.751e+00  -0.787  0.43120
## genreJazz         4.795e-01  2.932e+00   0.164  0.87014
## genreOther        -1.873e+00  2.329e+00  -0.804  0.42133
## genrePop          -1.092e+00  2.295e+00  -0.476  0.63416
## genreRock         -8.983e-01  2.346e+00  -0.383  0.70184
## energy            1.182e-02  2.016e-02   0.586  0.55775
## danceability      -3.883e-02  2.153e-02  -1.803  0.07165
## happiness         1.270e-02  1.342e-02   0.946  0.34425
## bpm              -3.741e-03  1.020e-02  -0.367  0.71388
## loudness_db       4.239e-02  1.071e-01   0.396  0.69223
## length_sec        1.714e-03  5.632e-03   0.304  0.76091
## age:genreJazz     -2.004e-02  7.403e-02  -0.271  0.78671
## age:genreOther     4.622e-02  6.149e-02   0.752  0.45237
## age:genrePop      -1.034e-02  6.027e-02  -0.172  0.86385
## age:genreRock     -1.651e-03  6.219e-02  -0.027  0.97883
## age:energy        -8.336e-04  6.954e-04  -1.199  0.23091
## age:danceability   9.308e-04  6.870e-04   1.355  0.17574
## age:happiness      1.533e-04  4.417e-04   0.347  0.72865
## age:bpm           7.048e-05  3.299e-04   0.214  0.83087
## age:loudness_db   -1.306e-03  3.605e-03  -0.362  0.71717
## age:length_sec    -4.083e-05  1.896e-04  -0.215  0.82959
## raceWhite:genreJazz 1.658e+00  1.996e+00   0.831  0.40631
## raceWhite:genreOther 9.835e-01  1.363e+00   0.721  0.47084
## raceWhite:genrePop  1.669e+00  1.355e+00   1.232  0.21811
## raceWhite:genreRock 1.991e+00  1.384e+00   1.439  0.15043
## raceWhite:energy    7.483e-04  9.274e-03   0.081  0.93570
## raceWhite:danceability -3.031e-03  8.984e-03  -0.337  0.73587
## raceWhite:happiness -7.637e-03  5.873e-03  -1.300  0.19377
## raceWhite:bpm      -3.436e-04  4.386e-03  -0.078  0.93758
## raceWhite:loudness_db 8.519e-02  4.553e-02   1.871  0.06158
## raceWhite:length_sec 4.248e-03  2.388e-03   1.778  0.07561
## genderNot Male:genreJazz NA         NA         NA         NA
## genderNot Male:genreOther -1.514e+00  8.832e-01  -1.714  0.08683
## genderNot Male:genrePop -1.077e+00  8.687e-01  -1.240  0.21536
## genderNot Male:genreRock -1.739e+00  9.057e-01  -1.920  0.05517
```

```
## genderNot Male:energy      4.152e-03  9.688e-03   0.429  0.66831
## genderNot Male:danceability 1.616e-02  9.138e-03   1.768  0.07734 .
## genderNot Male:happiness    -7.980e-03  6.116e-03  -1.305  0.19225
## genderNot Male:bpm          3.348e-03  4.539e-03   0.738  0.46084
## genderNot Male:loudness_db  -5.966e-03  4.702e-02  -0.127  0.89907
## genderNot Male:length_sec   -3.146e-04  2.412e-03  -0.130  0.89627
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.764 on 1111 degrees of freedom
## Multiple R-squared:  0.09126,    Adjusted R-squared:  0.05691
## F-statistic: 2.657 on 42 and 1111 DF,  p-value: 9.427e-08
```

```
step(model_total_2, direction="backward")
```

```
## Start:  AIC=1351.99
## rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + +gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##
##              Df Sum of Sq  RSS    AIC
## - race:bpm      1      0.019 3456.6 1350.0
## - race:energy    1      0.020 3456.6 1350.0
## - gender:loudness_db 1      0.050 3456.6 1350.0
## - gender:length_sec 1      0.053 3456.6 1350.0
## - age:bpm        1      0.142 3456.7 1350.0
## - age:length_sec  1      0.144 3456.7 1350.0
## - race:danceability 1      0.354 3456.9 1350.1
## - age:happiness   1      0.375 3456.9 1350.1
## - age:loudness_db 1      0.408 3457.0 1350.1
## - gender:energy    1      0.571 3457.1 1350.2
## - gender:bpm       1      1.693 3458.3 1350.5
## - age:energy       1      4.470 3461.0 1351.5
## - race:happiness   1      5.260 3461.8 1351.7
## - gender:happiness 1      5.296 3461.9 1351.8
## - age:danceability 1      5.711 3462.3 1351.9
## <none>                      3456.6 1352.0
## - gender:genre     3     19.903 3476.5 1352.6
## - age:genre        4     26.831 3483.4 1352.9
## - gender:danceability 1      9.725 3466.3 1353.2
## - race:length_sec   1      9.840 3466.4 1353.3
## - race:loudness_db  1     10.894 3467.5 1353.6
## - race:genre       4     33.047 3489.6 1355.0
##
## Step:  AIC=1349.99
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:loudness_db + race:length_sec +
##      gender:genre + gender:energy + gender:danceability + gender:happiness +
```

```

##      gender:bpm + gender:loudness_db + gender:length_sec
##
##      Df Sum of Sq  RSS   AIC
## - race:energy      1    0.015 3456.6 1348.0
## - gender:loudness_db 1    0.047 3456.6 1348.0
## - gender:length_sec 1    0.051 3456.6 1348.0
## - age:bpm          1    0.139 3456.7 1348.0
## - age:length_sec   1    0.146 3456.7 1348.0
## - race:danceability 1    0.350 3456.9 1348.1
## - age:happiness     1    0.380 3457.0 1348.1
## - age:loudness_db   1    0.408 3457.0 1348.1
## - gender:energy     1    0.561 3457.2 1348.2
## - gender:bpm        1    1.771 3458.4 1348.6
## - age:energy        1    4.462 3461.1 1349.5
## - race:happiness    1    5.242 3461.8 1349.7
## - gender:happiness  1    5.285 3461.9 1349.8
## - age:danceability  1    5.702 3462.3 1349.9
## <none>                3456.6 1350.0
## - gender:genre      3   19.886 3476.5 1350.6
## - age:genre         4   26.813 3483.4 1350.9
## - gender:danceability 1    9.741 3466.3 1351.2
## - race:length_sec   1    9.888 3466.5 1351.3
## - race:loudness_db  1   10.905 3467.5 1351.6
## - race:genre        4   33.031 3489.6 1353.0
##
## Step:  AIC=1348
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:danceability +
##      race:happiness + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##      Df Sum of Sq  RSS   AIC
## - gender:loudness_db 1    0.043 3456.6 1346.0
## - gender:length_sec  1    0.050 3456.7 1346.0
## - age:bpm           1    0.140 3456.7 1346.0
## - age:length_sec    1    0.151 3456.8 1346.0
## - race:danceability 1    0.361 3457.0 1346.1
## - age:happiness     1    0.372 3457.0 1346.1
## - age:loudness_db   1    0.430 3457.0 1346.1
## - gender:energy     1    0.547 3457.2 1346.2
## - gender:bpm        1    1.793 3458.4 1346.6
## - age:energy        1    4.460 3461.1 1347.5
## - gender:happiness  1    5.270 3461.9 1347.8
## - age:danceability  1    5.730 3462.3 1347.9
## <none>                3456.6 1348.0
## - race:happiness    1    6.582 3463.2 1348.2
## - gender:genre      3   19.874 3476.5 1348.6
## - age:genre         4   26.797 3483.4 1348.9
## - gender:danceability 1    9.732 3466.3 1349.2
## - race:length_sec   1   10.232 3466.8 1349.4
## - race:genre        4   33.016 3489.6 1351.0

```

```

## - race:loudness_db      1      21.451 3478.1 1353.1
##
## Step: AIC=1346.01
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:danceability +
##      race:happiness + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:length_sec      1      0.040 3456.7 1344.0
## - age:bpm                  1      0.135 3456.8 1344.1
## - age:length_sec          1      0.147 3456.8 1344.1
## - age:happiness           1      0.367 3457.0 1344.1
## - race:danceability       1      0.385 3457.0 1344.1
## - age:loudness_db        1      0.426 3457.1 1344.2
## - gender:energy           1      0.665 3457.3 1344.2
## - gender:bpm              1      1.819 3458.5 1344.6
## - age:energy              1      4.440 3461.1 1345.5
## - gender:happiness        1      5.591 3462.2 1345.9
## - age:danceability        1      5.716 3462.4 1345.9
## <none>                    3456.6 1346.0
## - race:happiness          1      6.565 3463.2 1346.2
## - gender:genre            3     19.858 3476.5 1346.6
## - age:genre                4     26.857 3483.5 1346.9
## - gender:danceability     1      9.746 3466.4 1347.3
## - race:length_sec         1     10.206 3466.9 1347.4
## - race:genre              4     33.010 3489.7 1349.0
## - race:loudness_db        1     21.859 3478.5 1351.3
##
## Step: AIC=1344.03
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:danceability +
##      race:happiness + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:bpm                  1      0.141 3456.8 1342.1
## - age:length_sec           1      0.159 3456.8 1342.1
## - age:happiness            1      0.365 3457.1 1342.2
## - race:danceability        1      0.397 3457.1 1342.2
## - age:loudness_db          1      0.430 3457.1 1342.2
## - gender:energy            1      0.634 3457.3 1342.2
## - gender:bpm               1      1.855 3458.5 1342.7
## - age:energy               1      4.419 3461.1 1343.5
## - age:danceability         1      5.722 3462.4 1343.9
## - gender:happiness         1      5.797 3462.5 1344.0
## <none>                    3456.7 1344.0
## - race:happiness          1      6.575 3463.3 1344.2

```

```

## - gender:genre          3      20.243 3476.9 1344.8
## - age:genre             4      26.971 3483.7 1345.0
## - gender:danceability   1       9.751 3466.4 1345.3
## - race:length_sec       1     10.488 3467.2 1345.5
## - race:genre            4      33.041 3489.7 1347.0
## - race:loudness_db      1     21.844 3478.5 1349.3
##
## Step:  AIC=1342.07
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:loudness_db +
##      age:length_sec + race:genre + race:danceability + race:happiness +
##      race:loudness_db + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:length_sec      1      0.129 3457.0 1340.1
## - age:happiness        1      0.309 3457.1 1340.2
## - race:danceability     1      0.389 3457.2 1340.2
## - age:loudness_db      1      0.455 3457.3 1340.2
## - gender:energy         1      0.612 3457.4 1340.3
## - gender:bpm            1      1.810 3458.6 1340.7
## - age:energy            1      4.280 3461.1 1341.5
## - age:danceability      1      5.768 3462.6 1342.0
## - gender:happiness      1      5.773 3462.6 1342.0
## <none>                  3456.8 1342.1
## - race:happiness        1      6.705 3463.5 1342.3
## - gender:genre          3     20.180 3477.0 1342.8
## - age:genre             4     27.117 3483.9 1343.1
## - gender:danceability    1      9.691 3466.5 1343.3
## - race:length_sec       1     10.558 3467.4 1343.6
## - race:genre            4     32.932 3489.8 1345.0
## - race:loudness_db      1     21.783 3478.6 1347.3
##
## Step:  AIC=1340.12
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:loudness_db +
##      race:genre + race:danceability + race:happiness + race:loudness_db +
##      race:length_sec + gender:genre + gender:energy + gender:danceability +
##      gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:loudness_db      1      0.399 3457.4 1338.2
## - race:danceability     1      0.427 3457.4 1338.3
## - gender:energy         1      0.631 3457.6 1338.3
## - age:happiness         1      0.745 3457.7 1338.4
## - gender:bpm            1      1.825 3458.8 1338.7
## - age:energy            1      5.119 3462.1 1339.8
## - gender:happiness      1      5.801 3462.8 1340.0
## - age:danceability      1      5.846 3462.8 1340.1
## <none>                  3457.0 1340.1
## - race:happiness        1      6.634 3463.6 1340.3
## - gender:genre          3     20.156 3477.1 1340.8

```

```

## - age:genre          4      26.990 3483.9 1341.1
## - gender:danceability 1       9.688 3466.6 1341.3
## - race:length_sec    1      10.489 3467.4 1341.6
## - race:genre         4      32.817 3489.8 1343.0
## - race:loudness_db   1      21.729 3478.7 1345.3
##
## Step:  AIC=1338.25
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + race:genre +
##      race:danceability + race:happiness + race:loudness_db + race:length_sec +
##      gender:genre + gender:energy + gender:danceability + gender:happiness +
##      gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability  1      0.391 3457.7 1336.4
## - gender:energy      1      0.687 3458.0 1336.5
## - age:happiness      1      1.160 3458.5 1336.6
## - gender:bpm         1      1.777 3459.1 1336.8
## - age:danceability   1      5.894 3463.3 1338.2
## - gender:happiness   1      5.905 3463.3 1338.2
## <none>                3457.4 1338.2
## - race:happiness     1      6.681 3464.0 1338.5
## - gender:genre       3     20.644 3478.0 1339.1
## - age:genre          4     26.895 3484.3 1339.2
## - gender:danceability 1      9.669 3467.0 1339.5
## - race:length_sec    1     10.443 3467.8 1339.7
## - age:energy         1     13.584 3470.9 1340.8
## - race:genre         4     32.662 3490.0 1341.1
## - race:loudness_db   1     21.920 3479.3 1343.5
##
## Step:  AIC=1336.38
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + race:genre +
##      race:happiness + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - gender:energy      1      0.608 3458.4 1334.6
## - age:happiness      1      1.118 3458.9 1334.8
## - gender:bpm         1      1.817 3459.6 1335.0
## - gender:happiness   1      5.781 3463.5 1336.3
## <none>                3457.7 1336.4
## - age:danceability   1      6.089 3463.8 1336.4
## - age:genre          4     26.770 3484.5 1337.3
## - gender:genre       3     20.857 3478.6 1337.3
## - race:length_sec    1     10.060 3467.8 1337.7
## - race:happiness     1     10.259 3468.0 1337.8
## - gender:danceability 1     10.419 3468.2 1337.8
## - age:energy         1     13.745 3471.5 1339.0
## - race:genre         4     36.911 3494.7 1340.6
## - race:loudness_db   1     21.534 3479.3 1341.5

```

```

##
## Step: AIC=1334.58
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + age:happiness + race:genre +
## race:happiness + race:loudness_db + race:length_sec + gender:genre +
## gender:danceability + gender:happiness + gender:bpm
##
##
## Df Sum of Sq RSS AIC
## - age:happiness 1 1.139 3459.5 1333.0
## - gender:bpm 1 2.219 3460.6 1333.3
## - gender:happiness 1 5.205 3463.6 1334.3
## <none> 3458.4 1334.6
## - age:danceability 1 6.050 3464.4 1334.6
## - gender:genre 3 20.559 3478.9 1335.4
## - age:genre 4 27.056 3485.4 1335.6
## - race:length_sec 1 10.061 3468.4 1335.9
## - race:happiness 1 10.123 3468.5 1336.0
## - gender:danceability 1 10.987 3469.3 1336.2
## - age:energy 1 13.985 3472.3 1337.2
## - race:genre 4 36.921 3495.3 1338.8
## - race:loudness_db 1 21.254 3479.6 1339.7
##
## Step: AIC=1332.96
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + race:genre + race:happiness +
## race:loudness_db + race:length_sec + gender:genre + gender:danceability +
## gender:happiness + gender:bpm
##
##
## Df Sum of Sq RSS AIC
## - gender:bpm 1 2.336 3461.8 1331.7
## - gender:happiness 1 5.618 3465.1 1332.8
## <none> 3459.5 1333.0
## - age:genre 4 26.019 3485.5 1333.6
## - gender:genre 3 20.285 3479.8 1333.7
## - age:danceability 1 9.624 3469.1 1334.2
## - race:length_sec 1 9.798 3469.3 1334.2
## - race:happiness 1 9.828 3469.3 1334.2
## - gender:danceability 1 11.054 3470.5 1334.6
## - age:energy 1 12.846 3472.3 1335.2
## - race:genre 4 36.800 3496.3 1337.2
## - race:loudness_db 1 21.219 3480.7 1338.0
##
## Step: AIC=1331.74
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + race:genre + race:happiness +
## race:loudness_db + race:length_sec + gender:genre + gender:danceability +
## gender:happiness
##
##
## Df Sum of Sq RSS AIC
## - bpm 1 0.277 3462.1 1329.8
## - gender:happiness 1 5.275 3467.1 1331.5

```

```

## <none> 3461.8 1331.7
## - gender:genre 3 20.237 3482.1 1332.5
## - age:genre 4 26.561 3488.4 1332.6
## - age:danceability 1 9.508 3471.3 1332.9
## - race:length_sec 1 9.748 3471.6 1333.0
## - race:happiness 1 9.754 3471.6 1333.0
## - gender:danceability 1 10.687 3472.5 1333.3
## - age:energy 1 13.542 3475.4 1334.2
## - race:genre 4 36.268 3498.1 1335.8
## - race:loudness_db 1 20.614 3482.4 1336.6
##
## Step: AIC=1329.83
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + loudness_db + length_sec + age:genre + age:energy +
## age:danceability + race:genre + race:happiness + race:loudness_db +
## race:length_sec + gender:genre + gender:danceability + gender:happiness
##
## Df Sum of Sq RSS AIC
## - gender:happiness 1 5.393 3467.5 1329.6
## <none> 3462.1 1329.8
## - gender:genre 3 20.130 3482.2 1330.5
## - age:genre 4 26.501 3488.6 1330.6
## - age:danceability 1 9.581 3471.7 1331.0
## - race:length_sec 1 9.695 3471.8 1331.1
## - race:happiness 1 9.796 3471.9 1331.1
## - gender:danceability 1 10.760 3472.9 1331.4
## - age:energy 1 13.719 3475.8 1332.4
## - race:genre 4 36.213 3498.3 1333.8
## - race:loudness_db 1 20.707 3482.8 1334.7
##
## Step: AIC=1329.63
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + loudness_db + length_sec + age:genre + age:energy +
## age:danceability + race:genre + race:happiness + race:loudness_db +
## race:length_sec + gender:genre + gender:danceability
##
## Df Sum of Sq RSS AIC
## <none> 3467.5 1329.6
## - gender:danceability 1 6.283 3473.8 1329.7
## - gender:genre 3 19.968 3487.5 1330.3
## - age:genre 4 26.600 3494.1 1330.5
## - race:happiness 1 8.978 3476.5 1330.6
## - race:length_sec 1 9.145 3476.6 1330.7
## - age:danceability 1 10.356 3477.9 1331.1
## - age:energy 1 13.092 3480.6 1332.0
## - race:genre 4 34.348 3501.8 1333.0
## - race:loudness_db 1 21.430 3488.9 1334.7
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
## danceability + happiness + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + race:genre + race:happiness +
## race:loudness_db + race:length_sec + gender:genre + gender:danceability,

```



```

##      data = cleaned_data)
##
## Coefficients:
##              (Intercept)                age
##              8.6220178                -0.0408272
##      genderNot Male                raceWhite
##              0.7280638                -1.4809661
##              genreJazz                genreOther
##              0.5285699                -1.6051711
##              genrePop                genreRock
##              -0.9281648                -0.6090139
##              energy                danceability
##              0.0159972                -0.0429294
##              happiness                loudness_db
##              0.0141879                0.0021139
##              length_sec                age:genreJazz
##              0.0006249                -0.0211649
##              age:genreOther                age:genrePop
##              0.0377222                -0.0147169
##              age:genreRock                age:energy
##              -0.0100985                -0.0009260
##              age:danceability                raceWhite:genreJazz
##              0.0010893                1.6839226
##              raceWhite:genreOther                raceWhite:genrePop
##              0.9821741                1.6598432
##              raceWhite:genreRock                raceWhite:happiness
##              1.9609966                -0.0076927
##              raceWhite:loudness_db                raceWhite:length_sec
##              0.0856656                0.0039525
##      genderNot Male:genreJazz    genderNot Male:genreOther
##              NA                -1.3988850
##      genderNot Male:genrePop    genderNot Male:genreRock
##              -1.0005892                -1.6516031
##      genderNot Male:danceability
##              0.0112198
anova(model_subpop1_1, model_subpop1_2)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + +gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##      Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      729 2510.6
## 2      700 2364.4 29    146.23 1.4928 0.0474 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

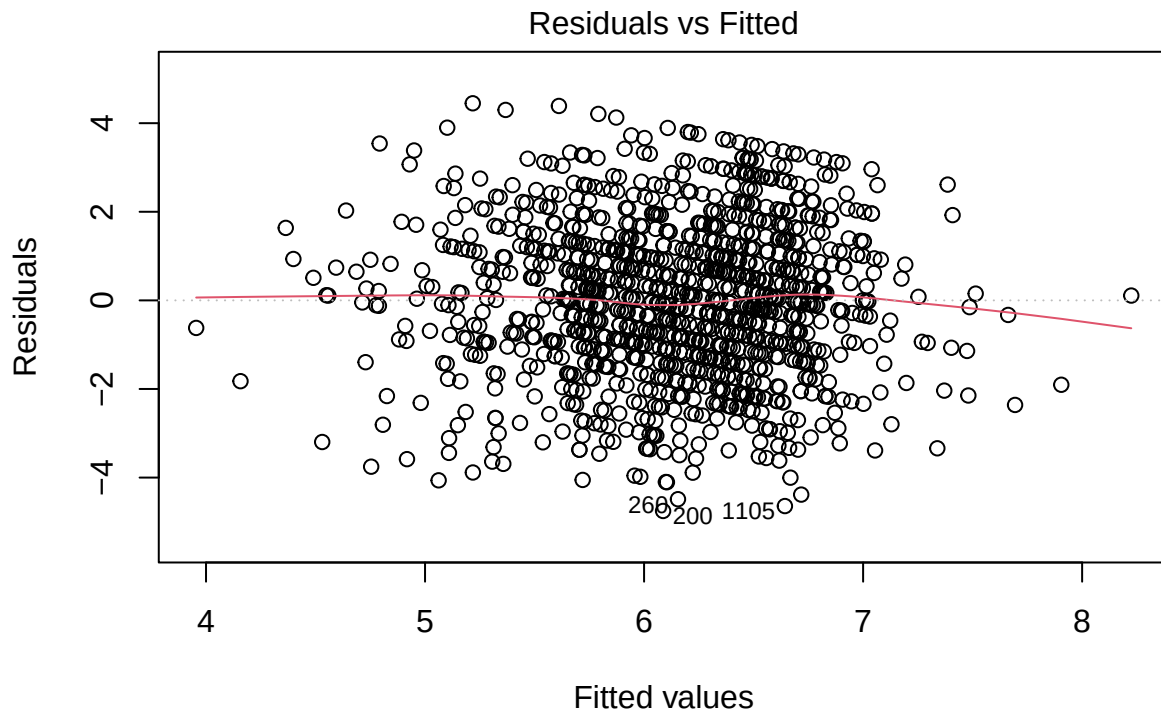
```
anova(model_subpop2_1, model_subpop2_2)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df      RSS Df Sum of Sq      F Pr(>F)
## 1      398 1012.58
## 2      374  919.86 24      92.717 1.5707 0.04418 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_total_1, model_total_2)
```

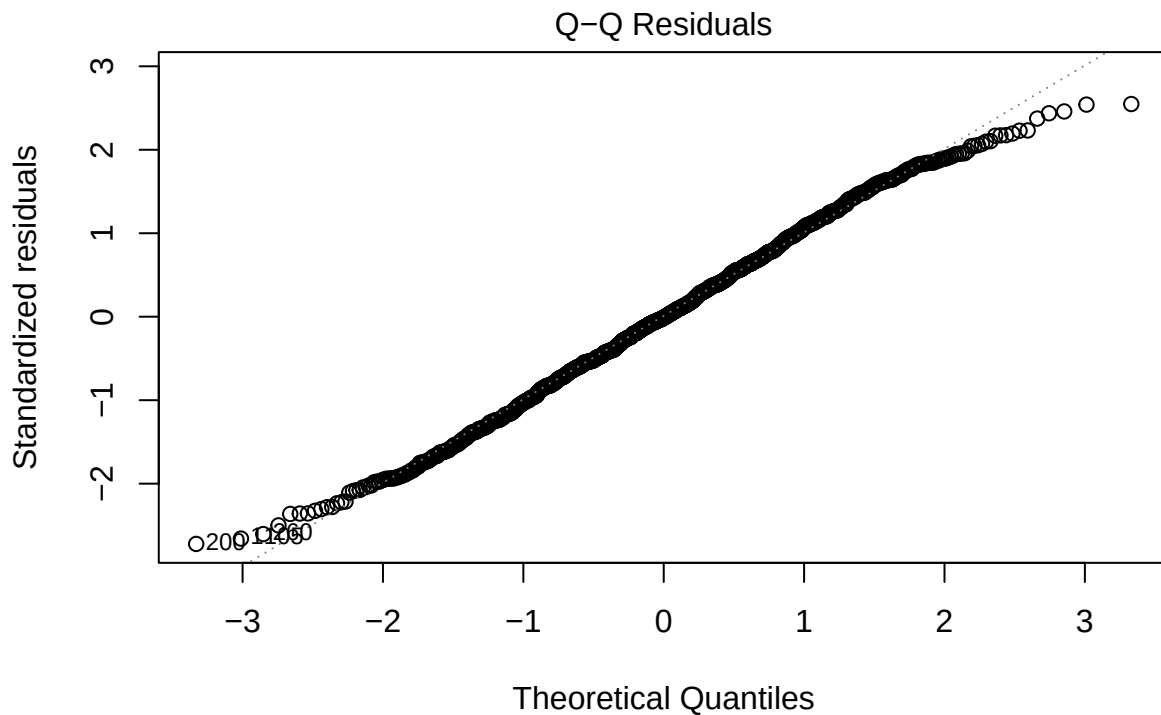
```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df      RSS Df Sum of Sq      F Pr(>F)
## 1      1140 3610.3
## 2      1111 3456.6 29      153.69 1.7034 0.01179 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot(model_total_2, which = 1)
```



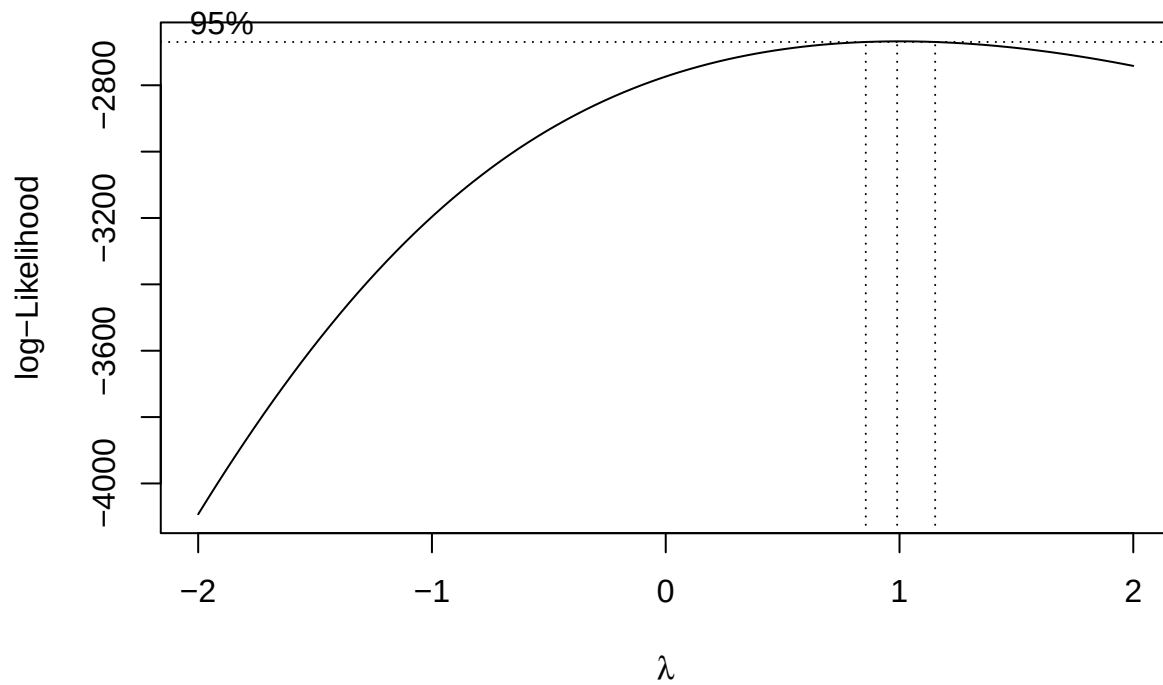
`lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...`

```
plot(model_total_2, which = 2)
```



`lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...`

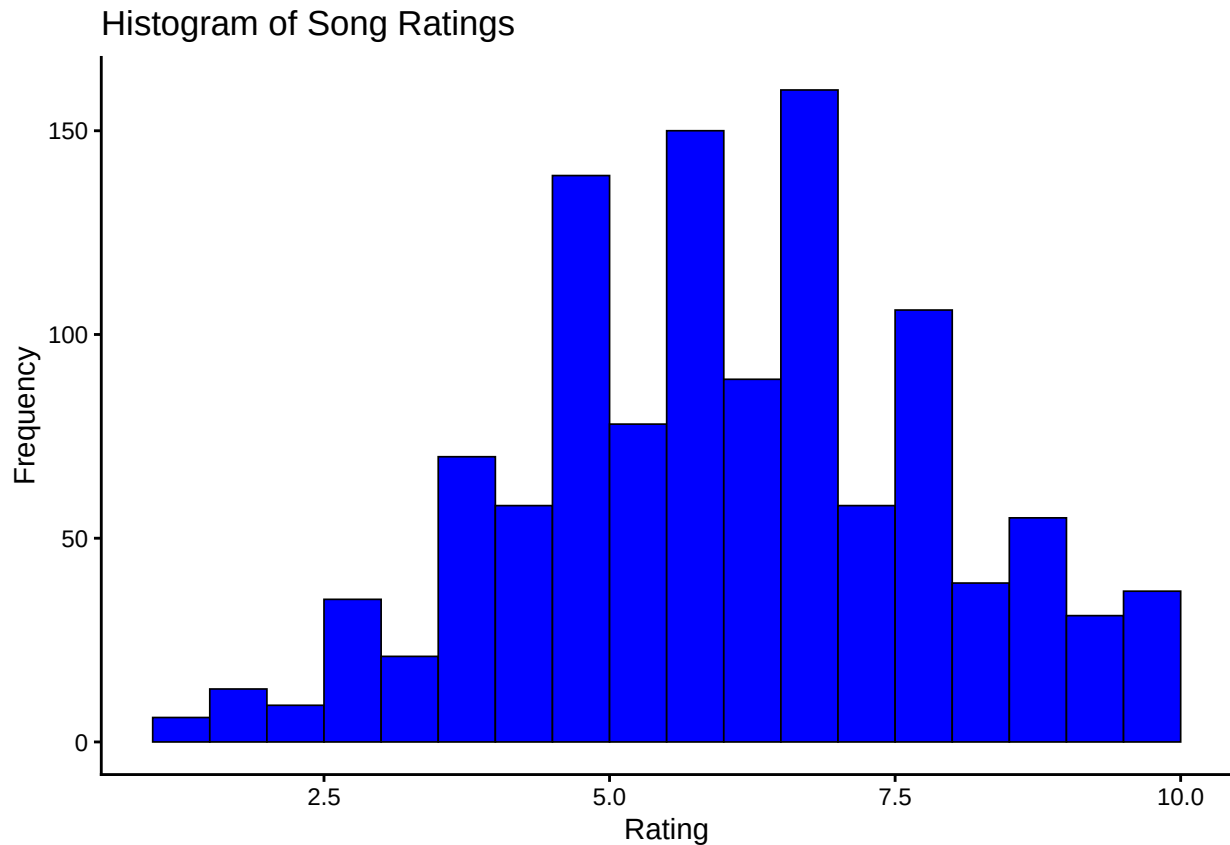
```
boxcox(model_total_2, plotit=T)
```



Preliminary exploration of data through visual analysis

```
library(ggplot2)

ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 0.5,
    boundary = 0.5,
    fill = "blue",
    color = "black",
    linewidth = 0.3
  ) +
  labs(
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()
```



The histogram of ratings appears approximately symmetric and unimodal, resembling a normal distribution. Although the ratings are discrete (integer-valued), the distribution spans multiple levels, creating a shape that approximates continuity. This supports the treating of rating as a continuous response variable in the regression analysis.

```
library(patchwork)
```

```
##
## Attaching package: 'patchwork'
## The following object is masked from 'package:MASS':
##
##      area

p1 <- ggplot(first_subpop, aes(x = age, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Age")

p2 <- ggplot(first_subpop, aes(x = energy, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Energy")

p3 <- ggplot(first_subpop, aes(x = danceability, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
```

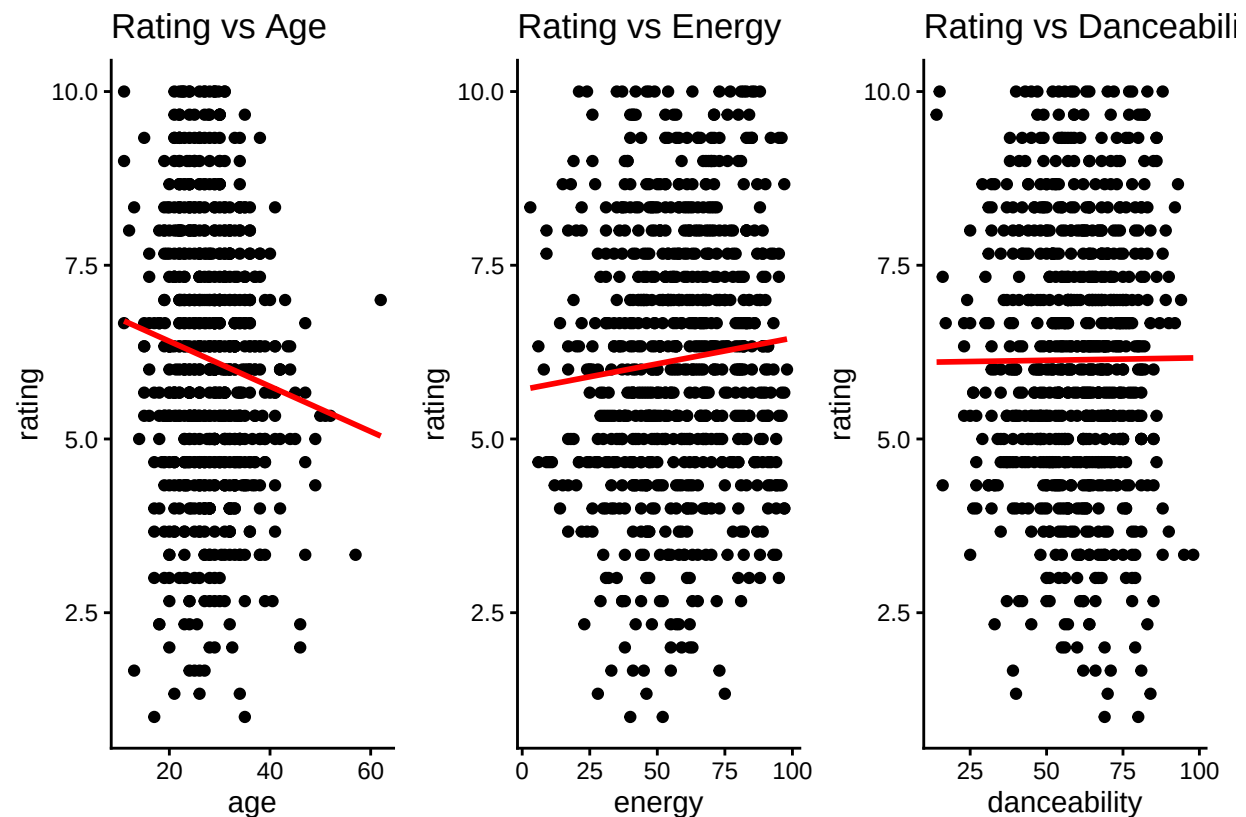
```
theme_classic() +
ggtitle("Rating vs Danceability")
```

```
p1 + p2 + p3
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



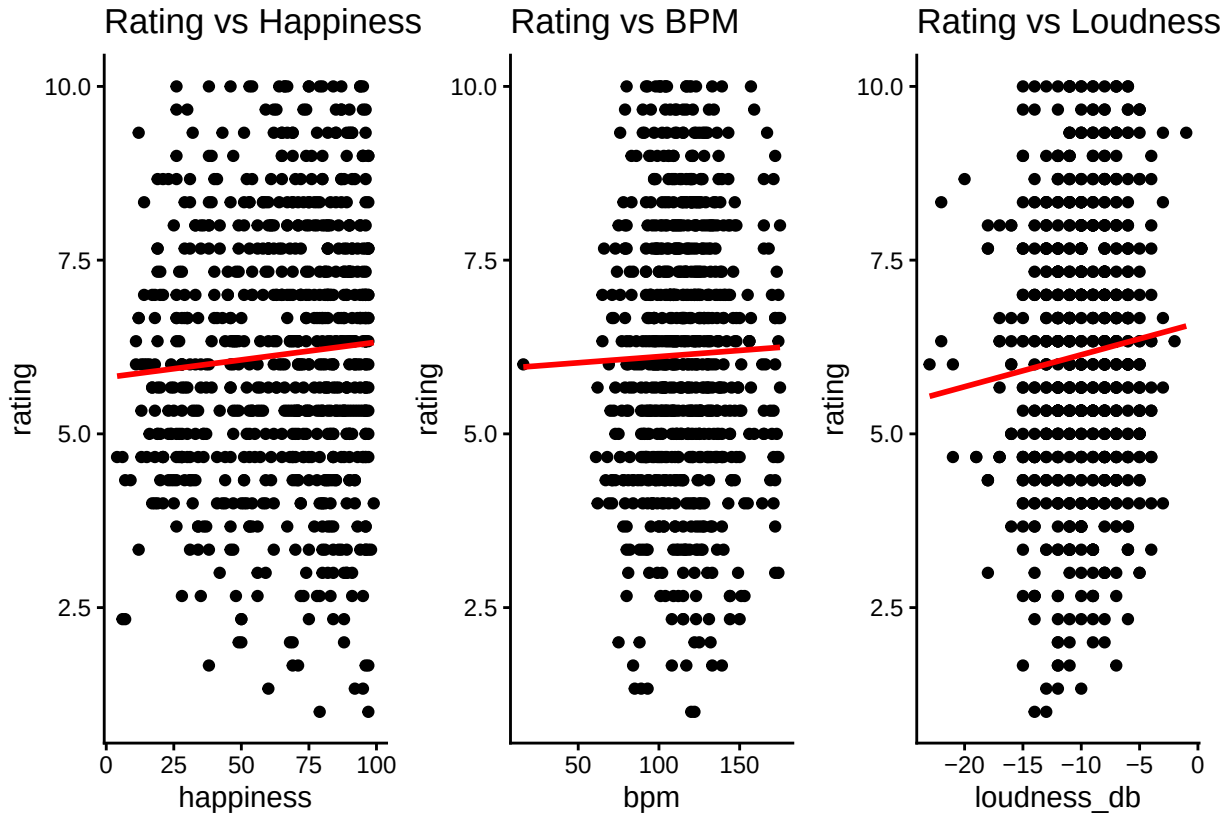
```
p4 <- ggplot(first_subpop, aes(x = happiness, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Happiness")
```

```
p5 <- ggplot(first_subpop, aes(x = bpm, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs BPM")
```

```
p6 <- ggplot(first_subpop, aes(x = loudness_db, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Loudness")
```

```
p4 + p5 + p6
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



```
p7 <- ggplot(first_subpop, aes(x = length_sec, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Song Length")
```

```
p7
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

